



ULTRA AVIATION SERVICES, INC.

General Operations Manual

SECTION J

AIRCRAFT

MOVEMENT & HANDLING

(AGM) (HDL)



Section: J Aircraft Movement & Handling

Revision: 7

Date: 02/02/2022

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J.1 RAMP SAFETY IN AIRCRAFT HANDLING

When conducting ground handling services, **SAFETY IS OUR TOP PRIORITY**. For this reason, it is essential that all company policies and procedures are followed while performing aircraft handling duties.

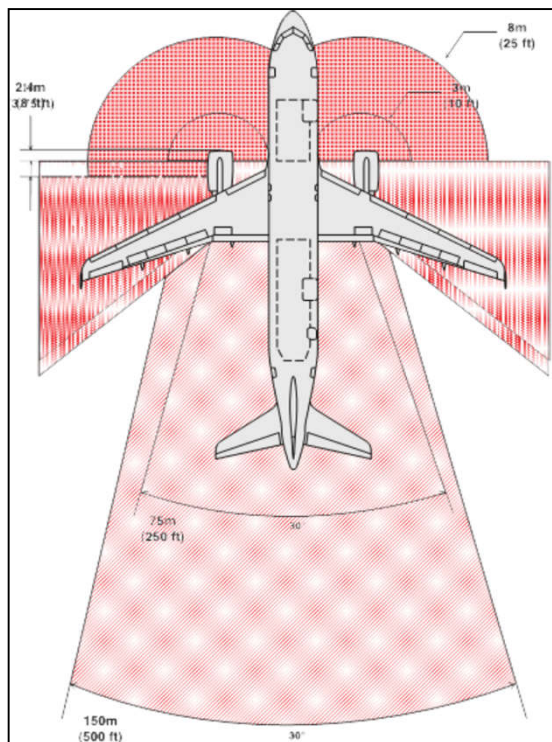
Aircraft damage can endanger passengers, employees and aircraft. Disruptions may also negatively impact safe airline operations. Even a slight scratch or dent on an aircraft may result in a serious accident.

If you see or cause any aircraft damage, you must report it immediately to your supervisor, the operations coordinator, and/or the safety manager.

J.1.1 DANGER AREAS

There is a particular risk of blast damage or injury from an aircraft engine's exhaust or intake. The risk is further increased if for any reason an aircraft tops and then applies the additional thrust required to "break away" and continue the maneuver.

Vehicles and personnel must remain clear of aircraft danger areas when the aircraft engines are running and/or the anti-collision lights are on. In order to prevent incidents and accidents caused by aircraft engines, never position yourself or any ground service equipment (GSE) in the engine Intake Area, Blast Area or propeller Rotation area (where applicable) during aircraft departure and arrival.



DANGER: Ground personnel and/or loose equipment **MUST STAY CLEAR** of the intake blast areas during arrival and departure.



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J.1.2 Equipment Restraint Area and Equipment Restraint Line (ERA)

HDL 1.2.8

Equipment Restrain Areas (ERA) is the area on the APRON (RAMP) where the aircraft is parked during ground operations. In some airports, such as MIA, they are indicated by a line. This area must be cleared of FOD and any GSE during the arrivals and departure operations. **GSE should not be driven faster than walking speed inside the ERA. Vehicles/ GSE not directly involved in the handling or servicing of the aircraft shall not be driven through or parked within the ERA.**



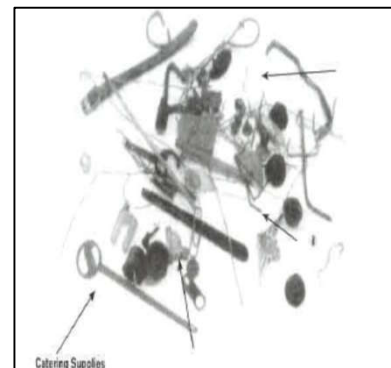
J.1.3 FOREIGN OBJECT DEBRIS (FOD)

AGM 1.1.1

Foreign Object Debris (FOD) is a general term which applies to all loose objects which are a danger to the safety and integrity of an aircraft and which **MUST NOT BE LEFT** in any area where they would constitute a hazard. Every individual has a responsibility to ensure that the risk of damage to aircraft from FOD is minimized. **ALL FOD MUST BE REMOVED AND PROPERLY DISPOSED OF** as soon as it is discovered.

Examples of FOD:

- Plastic and paper, bags/sheets, rags.
- Metal: Nuts and Bolts, Empty Oil and Hydraulic fluid cans, tools and equipment.
- Natural Objects: rocks, pebbles, and wood.
- Other Debris: Burst ballast bags, luggage handles and luggage wheels, etc.



CAUTION - Results of FOD:

Foreign object debris may be ingested into aircraft engines causing damage leading to engine failure. This is especially critical if it occurs in flight, particularly during the take-off phase.

In addition, damage caused by FOD can occur to tires, the undercarriage, control systems and other parts of the airframe. All such damage could lead to in-flight failures.



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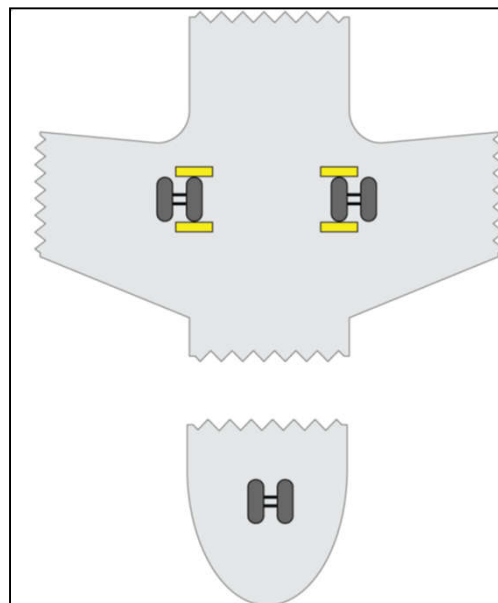
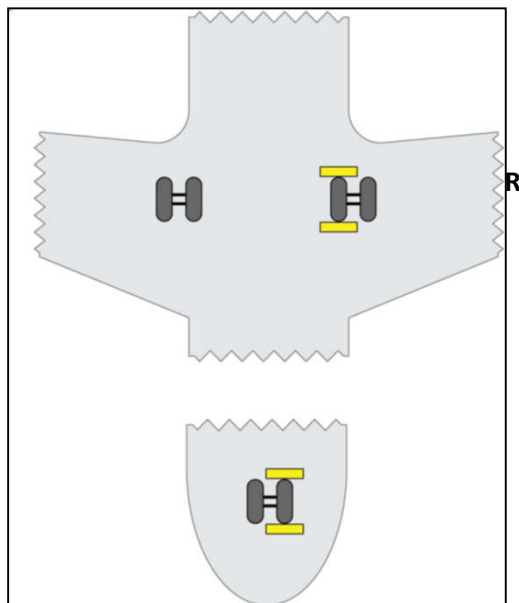
A. FOD CHECK PROCEDURE:

1. No garbage/ debris will be permitted in any piece of company own and or rented GSE
2. At ETA -10 all agents present at the gate area will perform an FOD walk around during which all pieces of debris, plastic, metal, natural will be picked up and disposed off inside the FOD barrel located at each aircraft parking location in the AOA or PLB.
3. During ground handling operations, all ramp agents will be alert to any piece of FOD that is generated so it can be pick up and disposed of immediately.
4. After termination of the GH operation crew will check gate area for any FOD, which needs to be picked up and disposed of accordingly.

J.1.4 WHEEL CHOCK PLACEMENT PROCEDURE

1. The company will use a minimum of four (4) wheel chocks during ground handling operations.
2. Chocks must be kept clear of the maneuvering area during aircraft arrival.
3. **DO NOT** approach the aircraft to position chocks until the aircraft has come to a complete stop.
4. Before approaching the main gear, make sure that the engines have been switched off and are spooling down and the anti-collision lights are switched off.
5. Walk towards the main gear in a path parallel to the fuselage, avoiding engine intake areas.
6. Place chocks according to carrier instructions and notify the flight deck that the chocks are inserted. See minimum placement Example below:

EXAMPLE:





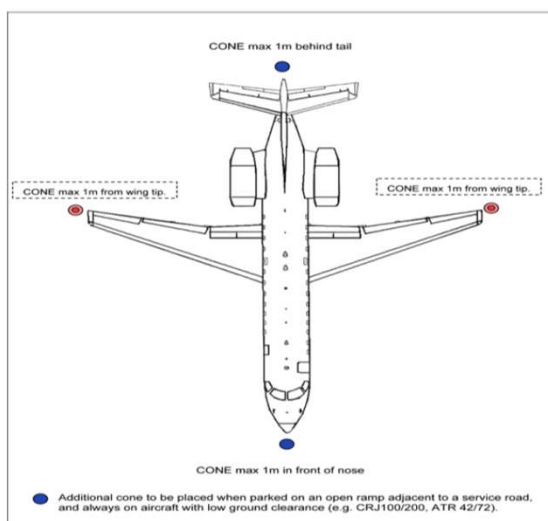
J.1.5 SAFETY CONES

HDL 1.3.7

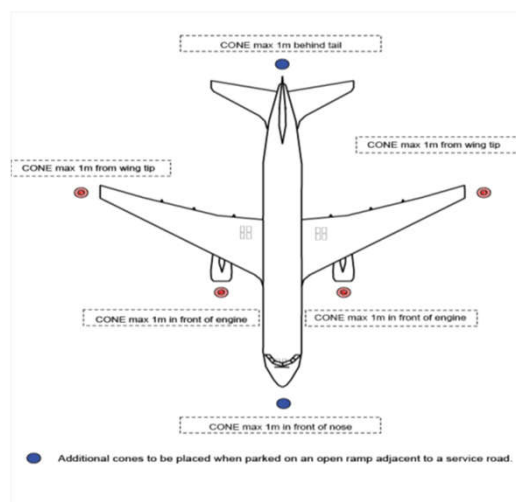
Safety cones are a caution sign for drivers to maintain required safety distances. Cones protect parts of the aircraft against collision by GSE.

A minimum of four (4) safety cones are typically used during ground handling operations. They are placed at the Nose, Tail, and each wing tip. Additional cones may be used according to client / carrier specifications. See below for placement examples.

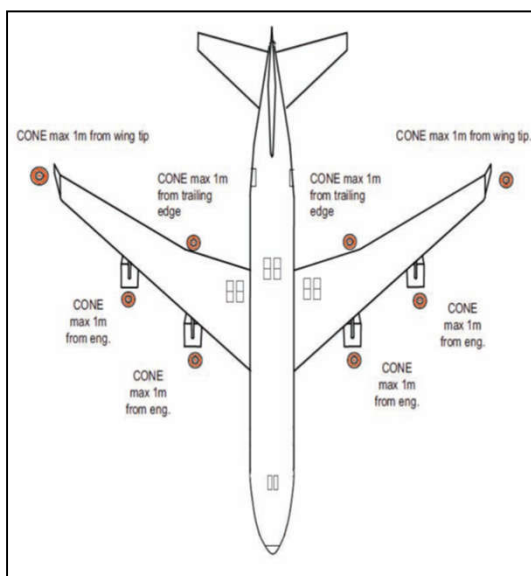
A. Minimum Cone Placement



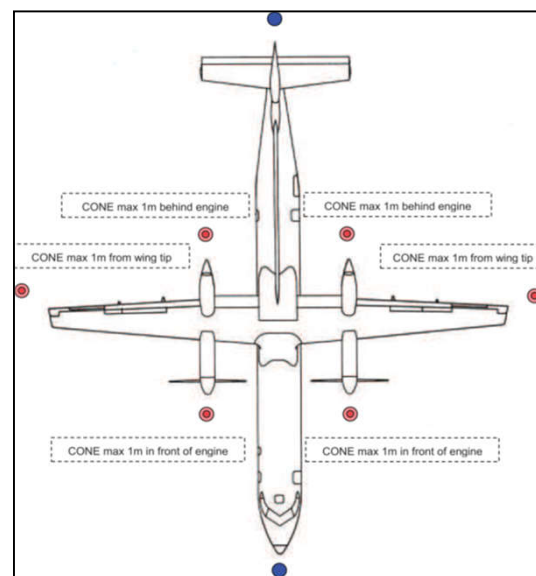
B. Cone Placement - Wing Mounted Engines



C. Wing Mounted / Four(4) Engine



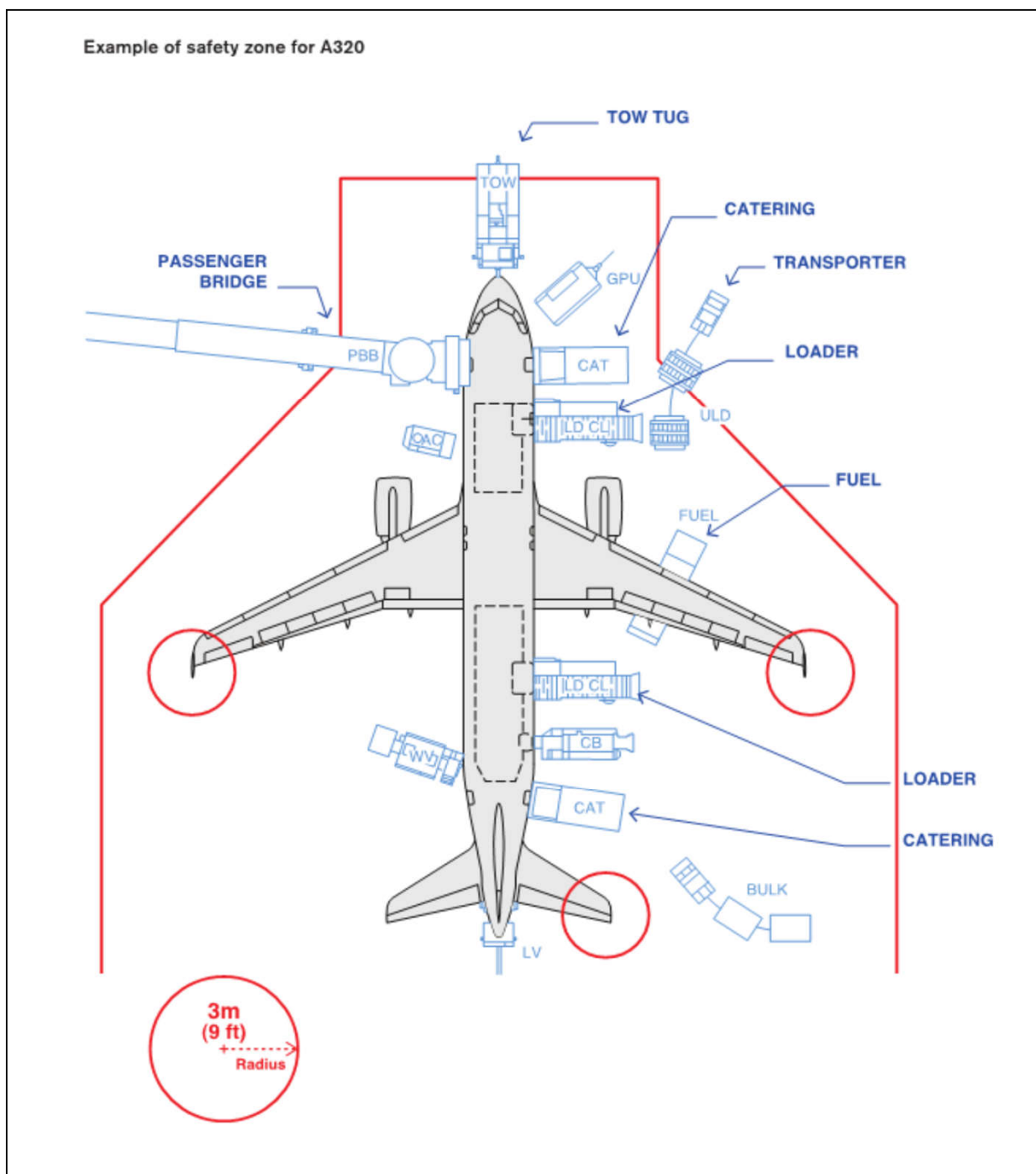
D. Cone Placement Wing Mounted (Propeller)





J.1.6 GROUND SERVICE EQUIPMENT (GSE) POSITIONING

During ground handling operations each piece of ground service equipment has its place. Please see the diagram below for standard GSE positions.





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Date: 10/01/2021**J.1.7 FUEL SAFETY ZONE****HDL 1.5.1**

For the protection of the aircraft and ground handling personnel in the ERA, a Fueling Safety Zone is defined as an area of at least 3 meters / 1- feet in any direction from the center-point of all fuel vent exits, refueling plugs, aircraft refueling ports, fuel hydrants, fuel hoses and fueling vehicles. This distance may be further increased as required by local airport or civil aviation regulations.

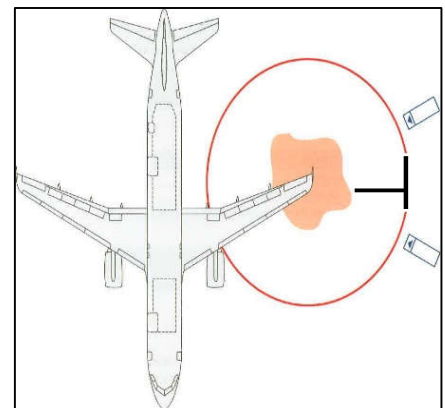
Within the Fueling Safety Zone, all personnel **MUST ENSURE** the following:

- **DO NOT SMOKE!**
- **ONLY USE** company issued and approved radios, radio telephones, torches, lamps and lighting systems. Battery Chargers must not be operated.
- Only enter the Fueling Safety Zone when required by present job task responsibility.
- **ALWAYS ASSUME THAT FUELING IS TAKING PLACE** anytime a fuel vehicle is on the stand during aircraft servicing and fuel hoses are connected.
- **DO NOT** leave vehicle engines running unnecessarily
- Position all GSE and vehicles so they do not obstruct the fueling vehicles' escape route. This is not a mandatory requirement for hydrant type fueling vehicles but every effort should be made to ensure a clear exit pathway.
- **DO NOT ALLOW** any passengers to enter the fueling safety zone.
- Avoid the use of motorized GSE within the fueling safety zone.
- Ensure fuel hoses are protected and all ground equipment is kept a minimum of 1 meter (3 ft) away from any fuel hose on the stand that is connected between a fuel truck and an aircraft.

J.1.8 FUEL SPILLAGE**HDL 1.5.1**

Take the following safety measures whenever a fuel spill occurs:

1. Activate the emergency shut-off valve where installed.
2. Alert the person in charge of fueling and/or the Pilot in Command of the spillage.
3. Contact the local fire service if not already done.
4. Verify with authorities/supervisor whether to stop all activity around the aircraft. Initiate aircraft evacuation, if necessary.
5. As far as possible, restrict all activities inside and outside the spill area to reduce the risk of ignition.
6. Secure the area 15 meters from the contaminated area.
7. Refer to the Spill Prevention Control and Countermeasures (SPCC) section of this manual for additional instructions.





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Date: 10/01/2021**J.1.8.1 AIRSIDE FIRE SAFETY****ORM 2.8.1**

Fire protection and prevention in ground operations conducted in station airside areas can have an immediate impact on airline and airport operations. A fire can leave equipment nonoperational, take an airplane out of service and may endanger ground crew, flight crew and passengers.

When entering the ERA to perform ground handling operations it is important to survey the area to identify and eliminate any conditions that may lead to a fire, and to check the availability and access to any firefighting equipment in the surrounding area, and of course **DO NOT SMOKE!** If you spot a potential fire hazard, please immediately report it to your immediate supervisor, or manager and the safety department.

Emergency Procedure - In case of fire:

- In case that an emergency presents fire in an aircraft in the process of loading fuel or any fire in the ramp, immediately advise the captain and crew onboard and the firemen.
- Initiate aircraft evacuation, if necessary.
- Use the Emergency cut valve if comes from a fuel vehicle.
- The switches are located on the fuel truck and on the wall of the terminal adjacent to the bridge area in some airports as applicable.
- Never use water in these types of fires because it would spread the fire.
- The ground personnel will inform immediately the firemen personnel and maintenance technician when a fire is discovered or presented on the landing gear.
- No one without the proper capacitation must fight the fire, because he/she can use the wrong extinguisher. And this can produce the tires to explode.
- After the fire suffocation, for safety measures assure that the tires and assemble are cold, to carry out any type of inspection.

Procedures for controlling and reporting fires

If you are directly involved in a situation of a fire while you are part of that operation, after the emergency was under control and in the next 24 hours a report must be made with details information to the Ultra Ops who will meet with the Airlines Customer when is required or Airport authorities.



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J.1.9 ADVERSE WEATHER CONDITIONS

Airside operational staff should follow these procedures during adverse or poor weather conditions which may have a negative impact on aircraft handling activities and ground safety. In the event that additional information is required, refer to supervisory staff.

A. WINTERY OR SLIPPERY APRON CONDITIONS

Winter weather brings extra hazards which require awareness and more care on the part of personnel working on the aprons to prevent accidents. The following precautions to reduce accident risk must be taken:

- **TAKE EXTRA CARE WHEN WALKING** across apron surfaces which can be slippery.
- **TAKE EXTRA CARE WHEN DRIVING**, especially when approaching an aircraft. Remember that vehicles may require additional distance to stop safely.
- Operators of potable water tankers and toilet servicing vehicles must be vigilant that there is no spillage or leakage that can lead to subsequent freezing.
- Close all entrance and cargo hold doors as soon as possible.
- **REDUCE SPEED** in slippery apron conditions.

B. THUNDERSTORMS AND LIGHTNING

Lightning strikes can severely injure or kill workers working outside during ground handling operations. The following notification levels and corresponding actions have been established for when lightning conditions occur.

LEVELS	ACTION
GREEN - NO ACTION - Possibility of Lightning or Thunderstorm activity OUTSIDE OF 5 MILES (8 km) of airport.	Disseminate Lightning Warning to airside operating staff.
AMBER - Lightning activity WITHIN 5 MILES (8 km) radius of airport.	Disseminate Lightning Warning to airside operating staff so they can prepare in case of Level 3. SECURE ALL ROLLING STOCK.
RED - STOP / SUSPEND - Lightning activity WITHIN 3 MILES (5 km) radius of the airport.	STOP ALL AIRSIDE ACTIVITY and SEEK SHELTER

- The local airport authority will also notify management staff when lightning conditions occur

**DANGER:** Failure to follow safety procedures could result in fatality!



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C. HIGH WIND CONDITIONS

High winds pose a great risk of damage and the following minimum precautions should be taken:

- Ensure the safety of the aircraft by installing additional chocks and removing all equipment from around the aircraft.
- Take extreme care when opening or closing aircraft hold doors .
- Make sure parking brakes are set on all parked GSE.
- Set parking brakes and secure by additional means if necessary, all non-motorized ramp equipment. (i.e. baggage carts and ULD dollies).

The following actions must be taken when sustained winds and/or gusts of wind exceed 25 knots are predicted:

ACTION	25-39 knots	40-59 knots	> 60 knots
Secure bag/freight carts, dollies, ladders/maintenance stands and tow bars and place near or against the building.	X	X	X
ENSURE parking brakes are set on all ground equipment.	X	X	X
ENSURE empty ULDs are secured and doors/curtains are closed.	X	X	X
CLEAR FOD and remove ULDs from the stands.	X	X	X
EMPTY FOD CONTAINERS and bring inside if not secure.	X	X	X
Suspend use of pre-conditioned air hoses and store securely. Remove marker cones.	X	X	X
ENSURE landing gear is chocked for high wind conditions.	X	X	X
DO NOT ELEVATE cabin service/catering highlifts and passenger stairs not equipped with stabilizers.	X	X	X
DO NOT ELEVATE cabin service/catering highlifts and passenger stairs not equipped with stabilizers.		X	X
CLOSE ALL cargo hold / passenger doors. Refer to Airline GOM.		X	X
CLOSE ALL aircraft access panels.		X	X
DO NOT ELEVATE booms on deicers.		X	X
REMOVE GSE FROM AIRCRAFT and secure in position outside of ERA clear of aircraft.		X	X
Secure boarding bridge and position to minimize surface exposed to the direct force of the wind.		X	X
Retract and lower boarding bridge. Position so that boarding bridge length points away from the wind.			X



TROPICAL STORM AND HURRICANE CONDITIONS: When Tropical Storm force winds are predicted be sure to follow all airport authority guidance and procedures. The airport will close once sustained winds reach 35 mph.



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J.1.10 SAFETY INSTRUCTIONS FOR OPERATING GROUND SERVICE EQUIPMENT (GSE) HDL 1.2.4, 1.2.5, 1.2.7, 1.2.8, 1.2.10, 1.10.1, 1.10.3

- **ONLY** drive or operate **GSE IF YOU ARE TRAINED** and authorized for that specific equipment type.
- GSE **MUST NOT BE** moved or driven across the path of taxiing aircraft or Embarking and disembarking passengers on the ramp.



- When Operating GSE, Check the contact zone for any possible aircraft damage and immediately report any damage found.
- Use All safety devices fitted on GSE (e.g. bumpers, handrails, stabilizers, etc.)
- Ensure protective rubber bumpers ARE NOT compressed against aircraft fuselage.

A. BASIC OPERATING REQUIREMENTS FOR GSE

1. Prior to initial use, operator must check the GSE assigned to them. (Brakes, Rubber Bumpers and Safety Systems.)
2. **DO NOT** carry extra personnel on GSE during movement without an approved seat. Always apply the **"NO SEAT NO RIDE"** principle
3. **DO NOT** operate GSE while using a hand-held portable electronic device.
4. GSE shall only be used for its intended purpose.
5. GSE shall never move across the path of a taxiing aircraft, or embarking and disembarking passengers.
6. **DO NOT DRIVE GSE** with lifting devices in the raised position. Except for final positioning onto the aircraft.
7. **DO NOT** operate GSE platform while in motion.
8. **DO NOT ALLOW ANY GSE** to move or be positioned **UNDER THE FUSELAGE** of the aircraft.
9. When GSE is unattended or positioned at or near the aircraft the parking brake shall be applied with the gear selector in "Park" or "Neutral" and, if equipped, wheel chock installed.
10. **DEPLOY STABILIZERS**, if equipped before operation.
11. **DO NOT TOUCH** the aircraft with any part of a GSE. Leave a 2 inch gap.
12. When positioning GSE ensure that a clearance is maintained between all GSE and the aircraft to allow vertical movement of the fuselage during the entire ground handling process.
13. All safety rails must be fully retracted/lowered during positioning and removal where possible.
14. **USE SAFETY RAILS** - Deploy all GSE safety rails when in use unless restricted by the aircraft.
15. Turn off the vehicle engine once the GSE is positioned, unless required for operation.
16. The Ground Power Unit (GPU) and Air Conditioning Unit may be left running unattended when connected to the aircraft, provided that fuel levels and serviceability are monitored.
17. **DO NOT** leave any vehicle unattended with its engine running.
18. **USE A GUIDE PERSON** when positioning GSE on an aircraft for service, when visibility is limited or when working in congested areas.
19. **MAKE SAFETY STOPS** when entering the ERA or at 5 meters when approaching an aircraft with GSE
20. **DO NOT** obstruct access to firefighting equipment or to the fuel hydrant emergency stop switch.
21. Securely stow GSE cables and hoses, where fitted, prior to transportation and when not in use.



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B. NON-MOTORIZED GSE**HDL 1.2.9**

In addition to the guidelines for motorized GSE, the following procedures for non-motorized GSE are as follows:

1. When parked, all non-motorized GSE **MUST HAVE BRAKES SET** or chocks in place when not connected to motorized vehicles.
2. ULDs must be secured on dollies (or trailers/trucks) using the appropriate restraints.
3. Pallet and container dollies may only be towed with the turntables in the locked position ("straight ahead") and rotated only when at the loader platform.
4. A **MAXIMUM OF 4** dollies or baggage carts or dollies may be towed at the same time.
NOTE: In the ERA, it is company policy that only two (2) dollies or carts should be towed into position by the aircraft for loading and offloading operations.
5. Avoid connecting non-motorized GSE to an aircraft and a tractor at the same time.
6. Before towing the unit away, make sure that the unit is disconnected from the aircraft and that
7. cables and hoses are securely stowed prior to transportation.



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C. GSE SAFETY INSPECTION**ORM 2.8.1**

Prior to putting any piece of Ground Service Equipment (GSE) in service a safety and functionality inspection must be performed. Check the following items for before placing operating a piece of GSE:

Motorized Equipment

- Current Airport Decal
- Fire Extinguisher
- Brake Lights
- Headlights
- Tail Lights
- Tire Condition
- Tire Pressure
- Fluid Leakage
- Rubber Bumpers
- Windshield / Windows
- Tow Hitch / Connector
- Controls / Steering
- Brake Test
- Engine Functionality
- Transmission Test
- Test Drive
- Safety Systems

Non-Motorized Equipment

- Tire Condition
- Tire Pressure
- Tow Hitch / Connector
- Stabilizers
- Locks / Curtains
- Equipment Functionality

Once the Ground Service Equipment has been inspected and is serviceable and in good condition, it can be put in use for ground handling operations.

D. UNSERVICEABLE GSE**ORM 7.1.1 (iii,iv)**

Ground Service Equipment with defective brakes, protective buffers (bumpers) steering tow couplings, lights, wheels or windshield wipers or with fuel or oil leaks must be tagged and removed from service and not utilized in airside operations.

Adopt the following procedures when unserviceable equipment needs to be removed from the operations area:

1. Tag the equipment as "Out of Service".
2. Notify the supervisor, who should contact the maintenance provider.
3. Remove the vehicle from the area according to the emergency breakdown procedures.
4. Do not operate vehicles or equipment labeled unserviceable until they have been repaired, inspected and the supervisor has released the equipment back into service.

J.1.11 ACCIDENT, INCIDENT, OCCURRENCE REPORTING

If an accident or incident occurs, **IMMEDIATELY NOTIFY** your supervisor, manager and/or the operations department. Refer to Section C of this manual for reporting and investigating procedures.



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Date: 10/01/2021**J.2 GROUND SERVICE EQUIPMENT (GSE) PROCEDURES****HDL 1.3.5**

Operating Ground Service equipment during ground handling operations carries with it the most potential for safety hazards, injury and/or damage to aircraft. It is vital that you remain alert and follow all safety procedures.

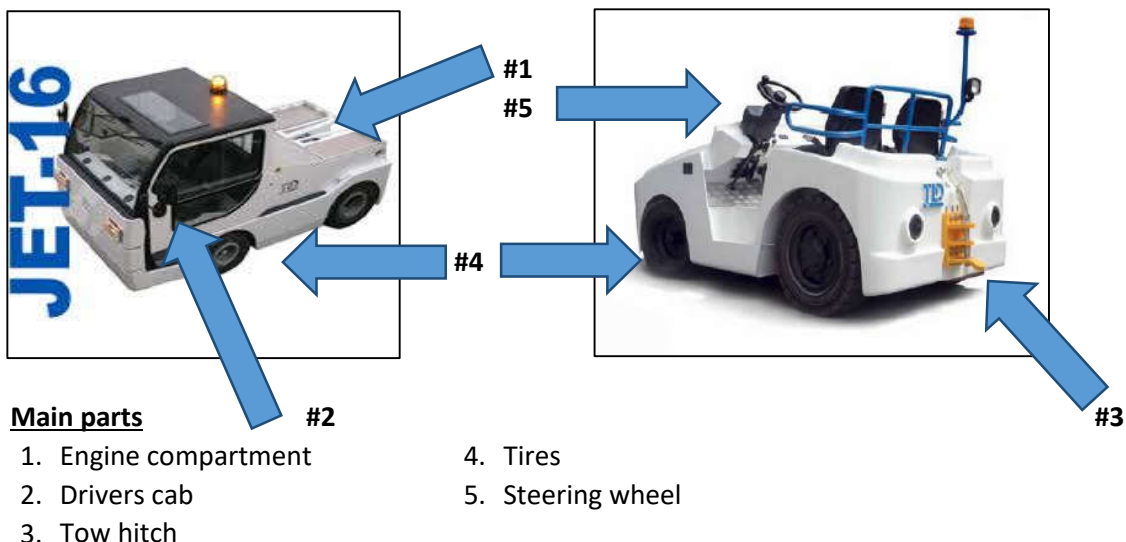
- **ALWAYS WEAR** your high visibility safety vest or approved reflective uniform when you are in the Aircraft Operating Area (AOA)
- **ALWAYS DISPLAY** your SIDA badge.
- **NEVER OPERATE** a piece of ground service equipment that you have not been trained and authorized to operate.
- **BE ALERT!**
- **NEVER DRIVE** lift enabled GSE in the elevated position.

This section contains the company procedure for operating Ground Service Equipment.

J.2.1 TUG / TRACTOR OPERATION**A. Requirements:**

- Valid Airport Badge with CBP seal
- Valid State of FL Driver's License
- Valid Airport AOA Driver's Permit (Yellow or Blue)
- Usage of required PPE I.e. Safety vest, Ear plugs, Company uniform, Safety shoes, the usage of Gloves is optional.
- Up-to-date certification on the equipment type

- B. Usage:** This type of GSE is a motorized tractor, which its mode of propulsion can be achieved either by internal combustion (gas or Diesel) power or electrical power. It is mainly used to move/ tow other type of non-motorized GSE from point A to point B for example, Ground Power Units (GPU), Air-Conditioning Units (ACU), Air Starter Units (ASU), baggage carts and dollies.





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D. TUG / TRACTOR OPERATION PROCEDURE



IMPORTANT: before attempting to operate any type of GSE make sure that you comply with all the requirements allocated to the operation of such equipment. (See Requirements).

1. **BEFORE** mounting / operating the equipment conduct a visual walk-around of the exterior and check for:
 - Broken Mirrors
 - Tire Pressure
 - Leaks
 - Broken Lights
 - Tow Hitch
2. Mount Tug / Tractor
Insert Key into ignition
3. Slightly turn ignition switch (key) until switch is "ON". **DO NOT TURN ALL THE WAY TO "START"**
4. Making sure that the parking brake is set. Dismount and turn light switch "ON". Check lights for functionality.
5. Check fuel or battery level
6. Mount equipment and fasten your seat belt
7. Pressing firmly on the brake pedal and with the parking brake set. Turn ignition switch to start.
8. Test horn
9. After releasing Parking Brake, conduct a transmission/ brake check, by selecting "Reverse" and allowing the tug to move backwards, apply brakes. Repeat test but now with the gear selector in "Forward".
10. Once the test are conducted if ALL of them are satisfactory then you can proceed with using the equipment. Remember for safety reason ONE single failed test can render the equipment "INOP".
11. Equipment with an "INOP or DO NOT OPERATE Tag" on them cannot be used
12. Ramp speed 20 mph or otherwise posted. 5mph inside the BMU.
13. **NOTE:** ON TLD electric tugs, you must be depressing the left pedal (third pedal) at all time for the tug transmission to operate. This particular type has a Gas pedal (right), a brake pedal (center) and the activator pedal (left).



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E. TUG / TRACTOR OPERATIONAL RULES

During operation of the equipment the following rules must be followed:

- **DO NOT** operate the equipment for any use other than for what it was intended.
- **DO NOT** use three ways hook ups. Tug- GSE- Aircraft.
- **ALWAYS LOOK BEFORE YOU MOVE**
- Make sure if towing a GPU, ACU, ASU that all your cables and hoses are stowed away properly.
- **DO NOT LEAVE TUG RUNNING UNATTENDED**, if you are going to dismount from it, engaged parking brake and turn ignition switch to "OFF".
- **DO NOT** tow more than 4 baggage carts or ULDs. Empty or otherwise.
- **DO NOT** exceed posted speed limits
- **USE A GUIDE PERSON** if operating in congested areas
- **NEVER LEAVE TUG RUNNING** or otherwise parked facing the aircraft. Parked it always parallel or facing away.
- Back the TUG up, when parking next to the building this will allow you a faster exit in case of any emergency
- **DO NOT** block fuel trucks or fire exits when parking or operating equipment
- When disconnecting baggage carts, GPU, ACU, ASU. Engaged parking brake on tug, Select "N" on gear selector, if driving a gas or diesel tug turn ignition to "OFF", on an electric tug moving your
- **NO SEAT, NO RIDE**
- **DO NOT** drive inside the circle of safety 6' (six feet) from aircraft fuselage out all the way around the aircraft
- Wait for the rotating beacon to be turned off before you approached any aircraft.
- Fasten your seat belt when operating driving the equipment
- Perform your daily checks. Do not drive/ operate any INOP unsafe equipment.
- **DO NOT** use your cellular telephone while driving
- **DO NOT SMOKE** within fifty feet of an aircraft or any refueling vehicle.



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J.2.2 BAGGAGE CARTS AND DOLLIES**HDL 1.3.6**

Baggage Carts and Dollies are used to temporarily store passenger baggage and cargo in transit between the baggage make up area and/or cargo warehouse and the aircraft. When using baggage carts and/or dollies use the following guidelines:

1. Before you hook up and tow a baggage cart(s) or dolly(s), perform a walk around to ensure that there are no chocks in place on any of the wheels and that the cart is serviceable.
 - If there are more than one carts or dollies connected together, check each connection and cart.
DO NOT CONNECT MORE THAN FOUR (4) CARTS OR DOLLIES together.
 - Take note of the turn radius required to clear any obstacles for the number of carts or dollies that you will be towing.
 - **SECURE THE LOAD**
 - If the cart(s) is(are) empty, open the curtain to ensure that no item has been left behind.
 - If the cart(s) is(are) full, close the curtain to ensure that no items will fall out during movement.
2. **RELEASE BRAKE** by dislodging the towing arm or depressing the hand brake.
3. Hook up the lead cart to the tug/tractor tow hitch.
4. **LOOK AROUND AND BLOW HORN** to ensure that no one is behind and/or between carts or dollies.
5. **DRIVE SAFELY** - When turning, take care to make turns which are wide enough to clear any obstacles.
6. Once you have reached your destination **DISMOUNT FROM THE TUG/TRACTOR** and disconnect the cart(s)/dolly(s) from the tug/tractor hitch
 - **DO NOT** disconnect carts by pulling a rope or other type of contraption adapted to the hitch.
7. **APPLY CART/DOLLIE PARKING BRAKE** by raising the tow arm until it locks in place. If the cart is equipped with a hand brake, then set it. If no braking system is present, chock one of the back
8. **DO NOT** allow any GSE to be positioned or parked under any portion of the aircraft fuselage
9. **DO NOT** leave GSE running unattended unless required by GSE operation.



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J.2.3 BELT LOADER**HDL 2.5.1, 2.5.2, 2.5.3**

The following precautions must be taken when operating a belt loader

1. Check that the belt loader is serviceable before use.
2. The boom of the belt loader should never be positioned inside the cargo hold of any aircraft.
3. Position and remove a belt loader in a straight line with the cargo hold door at a 90 degree angle to the aircraft fuselage .
4. Ensure the boom is clear of the aircraft or other obstacles before making a turn. If necessary USE A GUIDE to spot the approach.
5. The rubber bumpers on a conveyor belt loader **MUST NEVER MAKE CONTACT WITH THE AIRCRAFT**. Maintain clearance between the belt loader and the aircraft at all times.
6. **ALWAYS RAISE SIDE HANDRAILS** as soon as belt loader is positioned. Make sure they do not touch the aircraft fuselage .
7. Hand rails may be lowered to accommodate large items during loading and offloading.
8. **DO NOT** stand or walk on the belt when a hand rail is lowered.
9. Specially designed belt loaders (e.g. Ramp Snake or Power stow) require the equipment to be positioned inside the cargo hold.
10. **DO NOT** sit or stand on a conveyor belt while it is in operation (up or down)
11. When load/offload operation is complete. **SLOWLY** back the belt loader away from the aircraft an position outside of the ERA.

J.2.4 ULD LOADER**HDL 2.5.1, 2.5.2, 2.5.3**

The following precautions must be taken when operating a ULD loader

1. Check that the ULD loader is serviceable before use.
2. Check that the walking and loading surfaces are safe for use.
3. Lower both platforms during maneuvering of a ULD loader
4. The ULD loader must be outside the ERA before aircraft arrival and departure.
5. Make sure the movement path is clear before moving the ULD loader.
6. Never drive a ULD loader underneath the wing of an aircraft.
7. Move the ULD loader slowly towards the aircraft, avoiding any aircraft sensors or wing canoe fairings.
8. If visibility is limited or the aircraft type requires the ULD loader to be in close proximity to the fuselage or wing trailing edge, then a guide person must be used.
9. ULD loaders must never make contact with the aircraft. Position the ULD loader no closer than 2 in/5 cm or until the proximity sensors stop the movement (if equipped) .
10. **DO NOT** open/close aircraft cargo compartment doors while standing on a ULD loader. Use technical steps or a belt loader with a raised side safety rail, and deploy stabilizers if equipped. (Not applicable to main deck cargo doors).
11. Engage any safety systems and auto-leveler features if applicable. If the ULD loader is not equipped with an auto leveler, the level of the ULD loader must be monitored and adjusted as required.
12. When load/offload operation is complete. **SLOWLY** back the belt loader away from the aircraft an position outside of the ERA.



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J.2.5 PASSENGER STAIRS**HDL 1.2.6**

A Passenger Stairs Unit (PSU) is utilized when an aircraft is positioned at a hard stand where there is no loading bridge / jetway available. They may also be used as a secondary exit while fueling with passengers

A. Motorized Passenger Stairs Unit Procedures**HDL 1.4.1**

1. **BELT LOADER CHASSIS:** This PSU is mounted on a belt loader chassis with the difference that the belt control levers are used to operate stabilizers and stairs height.
 1. Before operation: Perform a safety walk around.
 - Check for any signs of fluid leaks
 - Check stair portion condition, and cleanliness
 - Check condition of tires
 - Remove wheel chocks
 2. Begin Operation: Turn ignition switch to "START".
 3. **TEST STAIR FUNCTIONALITY** by raising and lowering it. This is accomplished by "PULLING" on the lever to your right to raise and "PUSHING" on the same lever to lower.
 4. **TEST STABILIZER FUNCTIONALITY** by raising and lowering them. This is accomplished by "PULLING" on the right lever to raise and "PUSHING" on the lever to lower.
 5. Drive toward aircraft at a **SLOW SPEED** not to exceed, that of a person walking speed (3-5 mph)
 6. With the help of a guide person perform the final approach to aircraft L1 or L2 door which ever one you are using for deplaning/ embarking of passengers/ crew.
 7. Leave a two inches gap between the rubber bumper of the stair and the aircraft fuselage. **DO NOT COME IN CONTACT WITH AIRCRAFT FUSELAGE.**
 8. Adjust the height of the stairs.
 9. Apply the hand brake on the equipment.
 10. Lower stabilizers
 11. Go upstairs and give the Flight Attendant the thumbs up signal to open door thru the door window by knocking three times.
 12. Once door is open **ADJUST THE SIDE RAILS** to close gap between equipment platform and door.
 13. Turn equipment "OFF"
 14. **ENSURE** that the passenger stairs remain position at all times while the cabin door is open.
 15. After the last passenger embarks or disembarks and the "close door" signal is given by crew. RETRACT side rails and close door.
 16. REMOVE STAIRS once the cabin door is closed by an authorized person.
 17. Turn equipment "ON"
 18. Raise stabilizers
 19. Release parking brake from equipment and using a guide person reverse away from aircraft. **SPEED MUST NOT EXCEED THAT OF A PERSON WALKING (3-5 mph)**
 20. Park equipment behind equipment containment line.
 21. Set parking brake and lower stabilizers.
 22. Turn equipment OFF. Make sure that all light are turn off if the operation is at night and or during inclement weather.



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2. **TRUCK CHASSIS:** This PSU is mounted on a truck chassis, and drives just one. Operation is control for a control center inside the cabin. This control center is set to the right of the drivers and consist of as you are looking at it from left to right
1. Before operation: Perform a safety walk around.
 - Check for any signs of fluid leaks
 - Check condition of tires
 - Check stair portion condition, and cleanliness
 - Remove wheel chocks
 2. BEGIN OPERATION: Turn ignition switch to start.
 3. To test the control box you first will need to turn the master switch to ON and a red light will illuminate indicating that you have power.
 4. Moving the top toggle lock switch you will be able to release the lock for the A frame which will allow you to raise and or lower the frame to the desire inclination. NOTE you must hold
 5. Moving the lower toggle lock switch you will release the stair lock this will allow you to raise or lower the stairs to the desire height. NOTE you must hold this switch to unlock.
 6. Drive toward aircraft at a slow speed not to exceed , that of a person walking speed (3-5 mph)
 7. With the help of a guide person perform the final approach to aircraft L1 or L2 door which ever one you are using for deplaning/ embarking of passengers/ crew.
 8. Leave a two inches gap between the rubber bumper of the stair and the aircraft fuselage.
DO NOT COME IN CONTACT WITH AIRCRAFT FUSELAGE.
 9. Adjust high of stair
 10. Apply the hand brake on the equipment
 11. Lower stabilizers. This action will always move the stair forward about one inch, here is where point 7 becomes very important if not comply with it, you will scratch the fuselage
 12. Go upstairs and give the FA the thumbs up signal to open door thru the door wind on knocking three times. DO NOT OPEN DOOR UNLESS THE STAIRS ARE PROPERLY POSITIONED.
 13. Once door is open adjust side rails to close gap between equipment platform and door.
 14. Turn equipment "OFF". DO NOT TURN MASTER SWITCH OFF
 15. After the last passenger boards and the "close door" signal is given by crew. RETRACT side rails and close door.
 16. Turn equipment "ON"
 17. Raise stabilizers
 18. Release parking brake from equipment and using a guide person reverse away from aircraft.
SPEED MUST NOT EXCEED THAT OF A PERSON WALKING (3-5 mph)
 19. Park equipment behind equipment containment line.
 20. Lower stairs and A frame
 21. Lower stabilizers
 22. Turn Master switch OFF
 23. Set transmission lever to PARK
 24. Set equipment parking brake
 25. Turn OFF equipment.

CAUTION: Never drive the stair truck with the stair in the up position for a long distance, only to be positioned on the aircraft.



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B. PASSENGER STAIR UNIT NON-MOTORIZED

This type of Passenger Stair Unit is towed by a Tug/Tractor to the aircraft position. It will be positioned on the aircraft using man power.

1. Before operation: Perform a safety walk around.
 - Check for any signs of fluid leaks
 - Check stair portion condition, and cleanliness
 - Check condition of tires
 - Remove wheel chocks
2. Raise stabilizers by activating hand pump.
3. Push PSU to aircraft, you should always have a guide person when performing this task to continuously check the distance to aircraft.
4. Leave a two inches gap between the rubber bumper of the stair and the aircraft fuselage. **DO NOT COME IN CONTACT WITH AIRCRAFT FUSELAGE.**
5. Raise/ Lower stair, by rotating lever in middle of A frame counter clockwise to lower, clockwise to raise.
6. Lower stabilizers by activating hand pump.
7. Set chocks in front wheels
8. Go upstairs and give the Flight Attendant the thumbs up signal to open door thru the door wind on knocking three times. **DO NOT OPEN DOOR UNLESS STAIRS ARE PROPERLY POSITIONED.**
9. Once door is open adjust side rails to close gap between equipment platform and door.
10. After the last passenger boards and the "close door" signal is given by crew. **RETRACT** side rails and close door.
11. Remove wheel chocks
12. Raise stabilizers by activating hand pump.
13. Push PSU away from aircraft and park inside containment line
14. Lower stabilizers by activating hand pump
15. Set wheel chocks.



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J.2.6 GROUND POWER UNIT (GPU)

This auxiliary equipment is designed to provide to aircraft while on the ground.

1. Conduct Walk-Around Inspection

- Check electrical cable for signs of tear, exposed wires, any type of damage to the connection head plug.
- Check Tires
- Check Fuel Quantity
- Check Tow Tongue
- Check Manual Brakes

2. Position GPU by the front right side of the aircraft within the ERA and SET PARKING BRAKE and/or wheel chocks, if available.

3 Turn unit on at ETA -10 minutes, to allow engine to warm up

- a) Select toggle switch to "RUN"
- b) Select toggle switch to "START"

4. Once aircraft has been chock, Open GPU access panel

5. Pull electrical cord by the plug head. Extent to full length

6. Make sure that the six (6) female holes on the plug line up with the six (6) males prompts of the access panel.

7. Push the plug in until it cannot go any further

8 Close circuit by pushing green button on the GPU control panel

9. Once given the "OK" to disconnect electrical by the flight deck crew, Open electrical circuit by pushing in red button on GPU control panel. This action cuts the flow of electricity to the aircraft.

NOTE: DO NOT DISCONNECT CABLE UNLESS YOU HAVE FIRST CUT THE FLOW OF ELECTRICITY

10. Move toggle switch from RUN to IDLE

11. Disconnect electrical cable by softly pulling down the head.

12. Store electrical cable on it designated storage place

13. Turn engine OFF

14. Tow unit to designated parking position



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This auxiliary piece of equipment is designed to provide pre-cooled air to the aircraft while it is on the ground.

1. Conduct Walk-Around Inspection

- Check Fuel Quantity
- Check Tires
- Check Tow Tongue
- TEST ENGINE FUNCTION
- Check Manual Brakes
- Check air hose for signs of damage
- Check head plug for signs of damage.

2. When positioning ACU at the gate make sure it is within the ERA, and that you set the units parking brakes and chocks if available

Turn unit on at ETA +3 minutes, to allow engine to warm up.

- a) Select toggle switch to "RUN"
- b) Select toggle switch to "START"

3. Pull hose out completely, turn air lever to DOWN position to allow air flow and straighten air hose. Once accomplish return air lever to UP position to cut air flow**4. Connect nozzle to aircraft connection****5. Once connected softly pull down on hose to ensure a firm connection.****6. Open air flow by moving the lever to the down position****7. Once given the "OK" to disconnect air by the flight deck crew, turn air selector to neutral, cut flow of air by setting lever UP. NOTE: **DO NOT DISCONNECT HOSE UNLESS YOU HAVE CUT THE FLOW OF AIR******8. Disconnect hose by softly pulling down the nozzle.****9. Store air hose on it designated storage place****10. Turn engine OFF,****11. Tow unit to designated parking position**



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1. Conduct Walk-Around Inspection
 - Check Fuel Quantity
 - Check Tires
 - Check Tow Tongue
 - Check Manual Brakes
 - Check air hose for signs of damage
 - Check head plug for signs of damage.
2. Verify that only authorized personnel and equipment involved in the engine starting are permitted in the ERA during the engine start.
3. Establish communication with the flight crew to confirm the sequence and the number of engines to be started.
4. All personnel and equipment must remain clear from engine danger areas, including intake blast and propeller rotation, as applicable.
5. Position the ASU on the opposite side of the aircraft to the engine being started.
6. With the air level on the off position "Straight up" Turn unit engine on to ensure that it is working properly
7. When positioning ASU at the gate make sure it is within the ERA, and that you set the units parking brakes and chocks if available
8. Turn unit on at ETD -10 minutes, to allow engine to warm up.
 - a) Select toggle switch to "RUN"
 - b) Select toggle switch to "START"
 - c) Make sure that the air selector is in Neutral "Center" setting
9. Pull hose out completely , turn air lever to DOWN position to allow air flow and straighten air hose. Once accomplish return air lever to UP position to cut air flow
10. Connect nozzle to aircraft connection
11. Once connected softly pull down on hose to ensure a firm connection.
12. The headset operator informs the flight crew that the ground crew is ready for engine start. Once given the start up signal by the agent on the head set. Turn the air selector to JET START, right of center.
13. Start the engine(s) following the applicable requirements for the communication dialogue and signals.
14. Open air flow by moving the lever to the down position
NOTE: If unit is equip with a hand throttle pull on it until it is fully extended out. Otherwise unit will accelerated by itself
15. When engine start is complete and Once given the "OK" to disconnect air by the flight deck crew, turn air selector to neutral, cut flow of air by setting lever UP.
NOTE: DO NOT DISCONNECT HOSE UNLESS YOU HAVE FIRST CUT THE FLOW OF AIR
16. Disconnect hose by softly pulling down the nozzle. Make sure that you pull using the ring around the nozzle, otherwise you could get burn. Close and Latch external air start and electrical panels.
17. Store air hose on it designated storage place
18. Move toggle switch from RUN to IDLE
19. Turn engine OFF,
20. Tow unit to designated parking position



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J.2.9 GSE PARKING AND POSITIONING**HDL 1.3.2**

When Ground Service Equipment (**GSE**) **IS NOT IN USE** it must be parked in the following manner: GSE

- **MUST ONLY** be parked in designated airside parking areas outside of the ERA.
- **MUST NOT** obstruct access to firefighting equipment.
- **MUST NOT** obstruct access to the fuel hydrant emergency stop.
- **MUST HAVE PARKING BRAKE ON.**

J.2.10 GSE NEAR AIRCRAFT**HDL 1.2.7, 1.2.11, 1.2.12, 1.2.13**

When Approaching, Operating or Positioned near aircraft inside the ERA:

- **MUST NOT BE DRIVEN FASTER THAN WALKING SPEED.**
- When GSE is operated in close proximity to an aircraft, when the vision of the GSE operator is or might be restricted, ensure that any approach to the aircraft is guided by a company trained ramp
 - **Ensure that the guide person:**
 - i. Uses Hand Signals
 - ii. Is positioned so that clearance from the aircraft, other equipment vehicles or facilitates can be accurately judged, and signals can be visually communicated to the GSE operator.
 - iii. If visual contact with the guide person is lost, the GSE operator must stop movement of the GSE immediately.
- **ENSURE** that GSE is driven along a path that does not require sharp turns
- **APPROACH PARRALLEL** to the aircraft fuselage.
- **PARK GSE PARRALLEL** to the aircraft fuselage.
- When positioned near aircraft, **MUST HAVE PARKING BRAKE ON.**
- When positioned in operating mode near aircraft, the **OPERATOR MUST REMAIN** within easy reach of emergency controls, or in control of the equipment and **MUST NOT BE LEFT UNATTENDED** with the engine running except in cold weather operations with the GSE chocked.
- If fitted with stabilizers, **DEPLOY STABILIZERS**
- If fitted with auto-leveling, **DEPLOY AUTO-LEVELING**
- If fitted with hand rails, **DEPLOY HAND RAILS.**
- **ENSURE** that GSE attachment fittings, transfer bridges or platforms are correctly deployed
- **ENSURE** that rubber bumpers on GSE positioned at aircraft are not excessively compressed. **NO TOUCH POLICY.**
- GSE **MUST NOT** obstruct evacuation of persons from the aircraft in an emergency.
- GSE **MUST NOT** prevent or obstruct the movement of a fueling vehicle away from the aircraft.
- GSE **MUST NOT** impeded other aircraft handling operations in progress.



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J.2.11 POTABLE WATER SERVICING**HDL 1.5.3, 1.5.5****A. Pre-Cautions**

- Water service **MUST NOT** be performed by staff that has already performed toilet servicing during the same shift.
- Only uplift water if authorized by the operating airline.
- Replenish the aircraft tank according to the operating airline instructions.
- **HYGIENE PRECAUTIONS:** To perform water servicing you **MUST:**
 1. wear clean clothing and thoroughly wash your hands using soap before starting water service.
- **DO NOT:**
 1. Do Not fill the potable water service unit from the same water source as the toilet service unit.
 2. Do Not park the potable water service unit and the toilet service unit in the same area.

**IMPORTANT:** Maintain a distance of at least 100 ft (30 meters) between the water service unit and any lavatory service unit.

3. Do Not service the toilet and water on the aircraft at the same time.

Sequence: The water service should be performed prior to the lavatory service, however the carrier may prefer to perform the lavatory service first due to the available ground time and relative service times. If lavatory is performed prior to the water service, inspect the area for any signs of spill or contamination and wait 15 minutes before servicing the water after the toilet service has been performed.

B. Potable Water Servicing Procedure

1. Fill the aircraft water system as close to the departure time of the aircraft as possible.
2. Each aircraft type has specific requirements for filling and draining. Refer to the carrier's procedures for specific servicing instructions.
3. Open aircraft panel and wipe the aircraft filling port with an antiseptic wipe.
4. Before connecting the aircraft filling hose to the aircraft, flush the hose, by activating the pump and letting the water run for a few seconds.
5. Depressurize the system on the aircraft
6. Connect Hose
7. Turn the switch on the aircraft to "FILL" or to the desired water level quantity. (Check with the aircraft mechanic for desired water level.
8. Turn the switch the pump on the truck to the "ON" position, begin service.
9. When desired level is reached, end service by switching the pump to the "OFF" position.
10. Turn the aircraft panel switch to "OFF".
11. Disconnect hose and immediately cap the end to prevent bacterial contamination.
12. After allowing excess water to drain from aircraft filling port; cap the port.
13. Close aircraft panel
14. Check for any leakage and clean-up.
15. Safely pull away from aircraft

NOTE: When filling hoses are not in use, the nozzles or connectors must be protected from contamination either by the use of appropriate covers or by immersing them in receptacles containing chlorinated water. Do Not place the hose ends on the ground.



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C. Water Quality Standards**HDL 1.5.4**

This Procedure is applicable only to stations with a potable water truck or cart where the company conducts potable water servicing operations. The company will conduct a water quality test by use of a third party laboratory to ensure that there is no contamination and that the water quality is in accordance with local health authorities and those of the customer airline on a quarterly basis.

- If the water truck or cart is performing potable water servicing on a daily basis, then the vehicle must have a water quality test performed once every 31 days of use.
- If the vehicle is not used for several days, do not store water in the vehicle. Flush the water tank with approved chemicals.
- For vehicles which have been removed from service for maintenance for a long period of time, a water standards test must be performed and passed prior to returning the vehicle to service.
- This procedure does not apply to potable water obtained from approved potable water cabinets provided and maintained by the airport authority.
- The company will verify the credentials of the third party vendor who performs the water quality tests on an annual basis.

Water Testing Schedule

- **Water Truck** - Each water service unit will be tested on a bi-weekly basis by the company GSE Maintenance Department and a record of each test retained.
- **Airport Cabinet Test** - The company shall test the water source from which the water service unit is filled on a monthly basis by the company GSE Maintenance Department and a record of each test retained.
- **Laboratory Test** - Each water service unit will be tested on a quarterly basis and sent to a qualified laboratory for analysis. A record of the result shall be retained.



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J.2.12 TOILET SERVICING

The complete procedure for servicing the aircraft toilet waste tank consists of the following 3 steps:

1. Draining of the waste tank(s)
2. Flushing of the waste tank(s)
3. Adding an amount of pre-charge and/or a concentrated deodorant recharge product as applicable.

**CAUTION: TOILET FLUIDS ARE CORROSIVE.**

Prior to servicing, inspect the toilet servicing panel on the aircraft for signs of leakage. If any horizontal blue streaks are observed, the blue streak must be cleaned prior to servicing. After cleaning, look again for signs of leakage.

Blue Ice build-up in higher altitudes may influence airworthiness. In case of a possible leak, immediately inform the airline representative, ground engineer and/or advise the flight crew.

- **HYGIENE PRECAUTIONS**

1. Wear heavy rubber gloves, full face protection and protective clothing against harmful wastes when performing toilet servicing.
2. **DO NOT** park the toilet service unit in the same area as the water service unit nor at the water filling point. **Maintain a distance of 100 feet (30 meters)** between lavatory service unit and any water service unit.

A. TOILET SERVICING PROCEDURE

Each aircraft type has specific requirements for toilet servicing and the amount of recharge and/or concentrated deodorant precharge product. Refer to the operating airline's manuals for aircraft type specific instructions for more details.

1. Prior to opening a toilet service panel, check for stains around the panel.
2. While opening the service panel, stay clear and watch for signs of leakage.
3. Stay clear of the drain fitting cap while opening, and watch for signs of leakage.
4. Make sure the drain hose Y-fitting coupling is connected correctly, before a drain valve handle is pulled.
5. Empty the waste tank(s)
6. Flush the waste tank(s) twice and empty them again.
7. Precharge the tank(s) with the correct amount of water and disinfectant, as applicable.
8. Fill the waste tank(s) with the correct amount of water and concentrated deodorant precharge packets or pre-mixed fluid as applicable.
9. After servicing ensure that there are no leaks at the drain fitting cap and the end of the drain hose Y-fitting coupling.
10. Close the nozzle tightly in order to prevent the accumulation of ice during flight and wipe off residual water and disinfectant.
11. Check for possible leakage. NOTE: Inform aircraft maintenance or flight crew if fluid leakage is observed or the drain valve will not open or the waste tank cannot be drained.
12. After servicing close and latch the fitting caps and service panel door.
13. Empty the Toilet Service Unit, if necessary.



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B. AIRCRAFT DRAINING

1. Drain the aircraft waste system into the waste tank of a Toilet Service Unit.
2. Observe the waste drain hose during draining to confirm that the waste tank is completely emptied. The hose will also vibrate for a few seconds as the contents of the waste tank pass into the waste tank of a Toilet Service Unit.

Note: *Drain the waste tanks one at a time for optimal results*

C. INOPERATIVE TOILET SYSTEMS

If defects of the toilet system prevent regular servicing, ask qualified technical staff - if available - for assistance (e.g. removal of panels, etc.).

If no technical staff is available, inform the Flight Crew or an airline representative.

D. TOILET SERVICE UNIT DRAINING

Once the Toilet Service Unit is full, it must be emptied at the one of the airport tracherator facilities.

- **WIDE BODY:** Empty the toilet service unit after each servicing of a wide body aircraft.
- **NARROW BODY:** Empty the toilet service unit after 2-3 narrow body services, depending on the tank level.



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J.3 GROUND HANDLING PROCEDURES

This section contains general procedures for conducting below wing ground handling for client airline aircraft. Each client airline may have different procedures. Refer to the client airline's GOM. Client airline procedure will take precedence over these general procedures.

J.3.1 PERSONAL PROTECTION EQUIPMENT (PPE)**ORM 2.12.1**

Protective clothing and PPE provide a defense against operational hazards that could threaten the personal safety or health of ground handling personnel. Each agent assigned to perform functions in airside operations and in cargo areas should be equipped with the following personal protection equipment. Supervisors and Department Managers are responsible to ensure that ground personnel are wearing PPE according to company guidelines.

Required PPE:

- Safety Vest or Reflective Uniform Shirt
- Safety Shoes
- Gloves, Mask (Lavatory Truck)

Recommended PPE:

- Hearing Protection
- Knee Pads
- Lifting Belts



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J.3.2 AIRCRAFT CABIN DOORS

ORM 1.1.3

CABIN DOORS MUST ONLY BE OPENED BY TRAINED CREW in accordance with customer airline procedures. Passenger entrance and service doors are normally be opened from inside only by trained crew members. In exceptional cases cabin doors may be opened from outside. In such cases, special attention shall be paid to the disarming of the door-slider evacuation.

DO NOT ATTEMPT TO OPEN THE CABIN DOOR!



DANGER: Cabin access doors shall only be in open position if there is an appropriate boarding device positioned at the door. The boarding device must remain in position at all times when such door is open unless an appropriate fall prevention device is placed across the open door. Cabin access doors may not be opened without appropriate equipment positioned at the door. There is a risk of falling while operating cabin doors. Slide deployments can be fatal. If an armed door begins to open, do not attempt to hold the door as you risk being seriously injured or killed.

If a cabin access door is found open without a boarding device positioned at the door you must immediately notify a supervisor or the airline representative.

- **DO NOT** attempt to close the cabin access door unless trained and qualified.
- Guard the cabin access door until a qualified person is present to close it.

A. OPENING CABIN ACCESS DOORS FROM INSIDE BY TRAINED CREW

This is the only acceptable way for company employees to open passenger cabin doors.

Ground staff should:



CAUTION: Before allowing passengers or crew embarkation or disembarkation via a cabin access door, **ENSURE THAT THE BOARDING DEVICE IS PROPERLY POSITIONED AT THE DOOR**, and if stairs or integral air stairs are to be used, that both guard rails (if applicable) are extended. **See Passenger Loading Bridge and Passenger Stairs Procedures in this chapter.**

1. **KNOCK THREE TIMES** on the door from outside to indicate that a boarding device is properly positioned outside a door to be opened and that the door swing area is free of obstructions.
2. **STAND CLEAR** of the door and wait for the cabin crew to open.
(As applicable) Assist cabin crew with moving the door to the fully opened position and engaging the gust lock as necessary.



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B. RE-OPENING CABIN ACCESS DOORS**HDL 1.2.5**

If a cabin access door is not closed properly then it must be re-opened and re-closed.

Other situations when cabin access doors may need to be re-opened include the following:

- Subsequent delivery of catering and/or supplies, after the passenger boarding devices have been removed.
- Re-connecting of passenger boarding devices after the initial removal. If there is no crew on board the aircraft, call airline maintenance department for assistance.

Once the cabin access door has been closed in preparation for departure, do not attempt to re-open any aircraft door without the authorization of the flight crew.

- If you believe a door must be re-opened, you must notify the flight crew through an open cockpit window or use the flight interphone system.
- If the crew requires a door to be re-opened, they will notify ground staff. Regardless of which party requested that the door be re-opened, once the flight crew gives clearance for the door to be re-opened, follow the actions/steps in; Opening Cabin Access Doors.
- If authorization to re-open the door is not granted, do not attempt to re-open the door unless clearance given by the flight crew.



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C. CLOSING CABIN ACCESS DOORS

CABIN DOORS MUST ONLY BE OPERATED BY TRAINED CREW

in accordance with customer airline procedures.

NOTE: The passenger bridge should be removed immediately after the cabin door is closed.

1. Make sure service doors are closed immediately after servicing is completed.
2. Receive confirmation from the crew that the cabin access door(s) may be closed for departure.
3. Before removing the last boarding device from an aircraft, inform any ground staff onboard the aircraft that the last cabin access door is being closed and the last boarding device is being
4. Look for any possible obstructions around the door area and remove them.
5. Make sure the door gust lock is released and assist the person closing the door by moving it to the ajar position.



CAUTION: If the cabin access door cannot be closed with the boarding device connected, then the operation must be performed from inside the aircraft with extra vigilance and without assistance of ground staff outside the aircraft.

6. **DO NOT REMOVE** the boarding device from the aircraft until the door is fully closed and locked.
7. If stairs were used at a cabin access door, then retract the stair handrails if necessary to close the door. Remain at the top of the stair platform until the door is fully closed, and then descend the stairs before they are moved.
8. Close the door and carefully in accordance with the instructions and markings labeled on the door, and the respective aircraft type specific instructions.
9. Before leaving the vicinity of the door, confirm that the door is properly seated flush with the surrounding airframe and that the exterior door handle is flush with the surface of the door.
10. Seek assistance from aircraft maintenance personnel any time a door malfunction occurs.
11. **DO NOT** retract equipment stabilizers in advance of the cabin door being fully closed.
12. Before retracting equipment from the door, check to ensure the maneuvering area is clear of all obstructions and personnel.
13. If a passenger boarding stairs unit is used, then retract the passenger stairs canopy. Move the equipment to its approved parking position and engage any applicable restrains (such as closing the door on the passenger boarding stairs opening).
14. Visually inspect the cabin access door and the surrounding fuselage for signs of damage, particularly in any areas where the boarding device was in contact with the aircraft. If damage is discovered then immediately report it to aircraft maintenance personnel, and if available, the Pilot-in-Command.



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J.3.3 CARGO HOLD DOORS

Cargo hold doors are normally opened and closed by trained and authorized ground crew.

A. OPENING CARGO HOLD DOORS

1. **DO NOT** operate cargo doors unless trained and authorized.
2. Manual operation of an electrically or hydraulically operated cargo door may only be performed by maintenance personnel or flight crew.
3. **DO NOT** open the cargo doors until the aircraft engines have been shut down and the anti-collision lights have been switched off.
4. Before positioning loading equipment or any other ground support equipment at cargo doors and opening cargo doors, perform a visual check for any signs of damage to the doors or surrounding areas. If any irregularities are discovered during this visual check, report them to aircraft maintenance personnel and, if available, the Pilot-in- Command.
5. **USE TECHNICAL STEPS:** Cargo doors must be opened using technical steps or belt loaders equipped with raised safety rails to reach the cargo doors. ULD loaders must not be used. (Not applicable to main deck cargo doors).
6. Open the cargo doors in accordance with the respective aircraft type specific instructions.
7. Allow adequate space for door clearance to avoid equipment obstructing the free passage of the door:
 - i) Most aircraft lower compartment cargo doors hinge upwards. Be aware that when opening or closing cargo doors, the lower edge of the door will swing down before going upward.
 - ii) For main deck cargo compartment doors, remove safety barrier once the main deck loader is in position.
8. If the cargo door will not open, **DO NOT USE EXCESSIVE FORCE**, tools or ground support equipment to push or to pull on the door to open it. Contact aircraft maintenance personnel for assistance.



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B. CLOSING CARGO HOLD DOORS

1. Do not operate cargo doors unless you have first been trained and authorized.
2. Manual operation of an electrically or hydraulically operated cargo door may only be performed by maintenance personnel or flight crew:
3. Before closing the cargo doors, ensure; that load restraint and door protection nets are properly fitted.
4. That the cargo compartment lights have been switched off unless required for carriage of AVI.
5. That the door area including the door sill and frame are free of gravel, water, ice and other foreign substances or obstructions.
6. That the door and door frame show no visible signs of damage.
7. That any damage discovered during the inspection of the cargo doors and surrounding areas/frames is immediately reported to aircraft maintenance personnel and the Pilot-in-Command.
8. All cargo doors must be closed using technical steps or belt loaders equipped with raised safety rails to reach the cargo doors. ULD loaders must not be used. (Not applicable to main deck cargo doors).
9. Check that door lock indicators are engaged/properly set as applicable and that the door is properly locked, handles are stowed flush and panels are properly closed.
10. If a cargo compartment door is not closed properly, it must be re-opened and re-closed.



CAUTION: If a cargo door must be re-opened prior to aircraft movement, approval from the flight crew via the ground staff responsible for the departure must be obtained.

C. RE-OPENING OF CARGO HOLD DOORS

1. If a cargo compartment door is not closed properly, it must be re-opened and re-closed.
2. Once the pre-departure walk around has taken place, do not attempt to re-open any aircraft door without the authorization of the flight crew.
3. If you believe a door must be re-opened you must notify the flight crew through an open cockpit window or use the flight interphone system.
4. If the flight or cabin crew requires a door to be re-opened, they will notify ground staff.
5. Regardless of which party requested that the door be re-opened, if the flight crew gives clearance for the door to be re-opened, follow the actions/steps in: Opening Cabin Access Doors.
6. If authorization to re-open the door is not granted, do not attempt to re-open the door unless clearance is received from the flight crew.



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J.3.4 AIRCRAFT LOADING**HDL 1.8.1**

The loading of the aircraft will be according to the instructions of the client airline in a manner that prevents movement or spillage during flight. A load sheet will be provided to the Pilot-in Command for his final approval.

A. Supervision of Aircraft Loading**HDL 1.8.2**

Before loading, the hold shall be visually inspected for damage that can affect the load capacity. A qualified individual must supervise the loading of the aircraft and provide a signed confirmation to say:

1. The aircraft has been loaded as instructed – including any special load instructions;
2. The condition of locks, restraints or ULDs has not affected load capacity;
3. The bulk load and ULDs are correctly secured and locks and nets are in use;
4. Visible dangerous goods packages were inspected prior to loading;
5. Special loads, including dangerous goods have been stowed and secured according to regulations and operating airline procedures;
6. The holds are free of any foreign objects;
7. Any deviations are approved by the Load Agent and recorded;
8. At the completion of loading, a qualified individual must pass the final loading information to the Load Agent or Pilot-in Command to calculate the final weight and balance.
9. Ensure all loads, including visible dangerous good packages, are inspected for leakage and damage prior to loading.

B. Loading of General Baggage/ Cargo**HDL 1.8.8**

The flight coordinator or ramp supervisor is responsible for loading is in charge of the safe and efficient loading and offloading of the aircraft as well as the protection of the goods carried. He will ensure the aircraft is loaded as specified by the load agent, in accordance with the operating airline procedures.

NOTE: If the customer airline **DOES NOT CARRY** Dangerous Goods, Mail, or Cargo. The flight coordinator or ramp supervisor is responsible to **ENSURE** that such items are prevented from being loaded in the aircraft.

C. Loading of Dangerous Goods (where Carried)

The person responsible for loading is responsible for the loading of Dangerous Goods shipments as described in the IATA DG Manual and must be qualified as per IATA Dangerous Goods Regulations (DGR) Training requirements.

D. Special Precautions when using Carts and Dollies

1. Do not wedge light packages between heavier items.
2. During transportation in carts and dollies ensure that the load is properly secured by using appropriate locks, stops, rails, curtains and straps.
3. Ensure the overall height of load permits safe lifting of each piece of load during loading and offloading of carts by personnel standing on the ground.
4. When using tarpaulins, ensure all straps are securely fastened to the baggage cart.
5. When not in use the braking system shall be engaged on all strings of baggage carts.



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E. Safety Requirements Specific to Aircraft Loading Operations**ORM 8.2.1**

1. Get assistance when moving heavy articles.
2. Do not use baggage carts to gain access to cargo compartments.
3. The loader bridge height shall be monitored during the loading process and adjusted as necessary to maintain a correct alignment with the cargo hold floor.
4. Use ULD loader platform guide rails as required to ensure alignment when loading.
5. Block or secure cargo which will not lie flat on conveyor belts.
6. Push DO NOT PULL containers on and off dollies and loaders.
7. ULDs on dollies or transporters must be secured to prevent movement by the use of locks, stops, rails or straps at all times, except when the load is being transferred onto or off the equipment.
8. Protect live shipments from inclement weather.
9. Be alert for special/ dangerous goods shipments. Know how they must be handled and secure.
10. Operators of equipment shall ensure that other personnel are not entrapped by movement of load/ pallets/containers either in the aircraft or on the loading equipment.
11. Gates of loaded carts should be lowered carefully, in case loose cargo falls out and causes injury.
12. Holds and compartments shall only be entered or exited by using the appropriate elevating device and which has been positioned and secure, e.g. belt conveyor and cargo loader.
13. Elevating devices must not be removed from the aircraft when personnel are still within the cargo hold.
14. Do not walk between ULDs or carts being towed, or when they are stationary on the ramp as you do not know when they may be moved.
15. When loading has been completed, move all loading equipment well clear of the aircraft.

F. Special Requirements when using Tractors

1. Drive tractors and carts within speed limits according to local airport regulations, and take care to avoid sharp turns, jerks and sudden stops.
2. Approach the aircraft at walking speed.
3. Limit the number of carts and dollies in a train to the maximum specified by the local airport regulations.
4. Do not attempt sharp turns close to the aircraft. Keep at least 1 m (3 Ft) away from the fuselage.

G. Special Precautions when using Belt Loaders

1. Ensure proper separation between articles on the conveyor belt to avoid jamming.
2. Adjust the back of the conveyor belt correctly to avoid dropping goods from the belt.
3. Handrails shall be deployed when a belt loader is used to gain access to aircraft cargo holds; however caution shall be exercised where there is restricted clearance with the aircraft fuselage or engines.



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H. Special Precautions when using ULD Loaders

1. **DO NOT** stand between loader and dollies.
2. Align dollies correctly to loader. Use guide if required.
3. ULD loaders should not be used to transport ULDs across the ramp unless specifically designed for this purpose.
4. Do not move ULDs until personnel are clear of all hazards.
5. The vehicle (dolly or other) shall be positioned within 7.5 cm/ 3" of the loader platform periphery and shall be at the same height before transfer takes place.
6. The use of external mechanical force to move ULD should not be necessary except with very heavy ULD which may not move by manual force alone. Any such process shall be carried out with approved equipment and procedure.
7. Whenever possible, reverse in a straight line at a walking pace, monitoring all sides for clearance.
8. Ensure path of loader is clear of all obstructions prior to initiating turns.
9. Retract or lower handrails, platforms and operators sections to ensure adequate clearance before positioning equipment.
10. Extend all handrails before loading or unloading.
11. All personnel must keep clear of the elevating platform when it is being raised or lowered.
12. Do not rotate a ULD on a loader elevator platform while raised or in-transit.
13. Before removal or repositioning of loader, ensure any load positioned in the doorway is secured against roll out.



CAUTION: Warning Fall Hazard: Do not ride on the elevating platform to gain access to the loader bridge.

I. Low – Wing Aircraft

1. To prevent damage to aircraft with low wings, baggage tractors shall stop approximately 1 meter (3 Ft) from the belt loader to unhook the carts. Move the tractor away and position the carts by hand.
2. Take care when pulling or pushing carts especially when ramp conditions are slippery. When necessary obtain assistance.
3. When removing baggage carts, the tractor shall be positioned pointing away from the aircraft wing and baggage cart pulled to the tractor.
4. Extreme caution must be used when using covered baggage carts.

J. Bulk Loading of Smaller Aircraft

1. Use belts loaders if the door still height does not allow items to be passed into the doorway without undue lifting Always consider the use of belt loaders for heavy items (over 23KG/ 50lbs).
2. Keep a gap of at least 1 m (3 Ft) between baggage carts/ dollies and the cargo belt when towing, to prevent collisions when approaching the belt loader.
3. Carts or dollies must be disconnected from the tractor and maneuvered by hand if the carts or dollies need to be closer than 1 m (3Ft).



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1. **Hold baggage must be inspected for signs of leakage before loading.**
2. Any item of load which is not properly packed and any item that may damage or contaminate the aircraft must not be loaded and the customer airline must be notified.
3. Container curtains doors and/or nets must fully be closed and locked into place prior to loading the ULD into the aircraft.
4. Containers must not be contaminated when loaded (snow, wood, plastic etc.).
5. Use tarpaulins or covered carts during inclement weather.
6. Do not place goods directly on the apron.
7. Always observe the specific instruction labels and marks such as FRAGILE, TOP, THIS SIDE UP, etc.
8. Report torn (or missing) baggage tags and cargo labels, and do not load unless corrected.
9. Report immediately any damage to the load, whether it occurs during handling or is noticed on arrival.
10. Report immediately any spills, unusual fumes or smells, etc. to a Supervisor, Flight Crew or local authorities as required.

L. Spills in cargo Holds

1. Spills can occur in cargo holds during loading and in flight due to :
 - i) Improper packaging.
 - ii) Damage due to mishandling prior to loading.
 - iii) Improper loading in the compartment.
2. Spills can be liquid, gels, or material in a powdered or granulated form.
3. Spills can be hazardous corrosive, flammable, explosive, toxic or poisonous, etc. Even water can cause serious damage to electrical components and systems.
4. Spills can be corrosive to the aircraft structure. Mercury spills are particularly corrosive to the extent that the affected aircraft structure may have to be completely replaced if not cleaned up quickly.

IMPORTANT: It is essential that any spill is reported immediately to Maintenance, so that corrective action can be taken. Initiate the local emergency response plan for spill events.

M. Cargo Hold Inspection**HDL 1.8.3**

When and offload is completed, a final check of ALL cargo holds must be conducted to inspect each cargo hold:

1. Damage to the compartment.
2. Damaged or malfunctioning floor locks.
3. Spills in the hold that may have occurred.
4. Baggage or cargo that may have been left onboard the aircraft.
5. Any other items that should not be present in the hold.

NOTE: A check must be conducted in a hold even if on arrival the hold was reported as not carrying any cargo/baggage (empty). If any damage is found to the compartment or locks, if a spill has occurred, or if any other irregularity is found, it must be immediately reported to a supervisor, the flight crew, and/or company representative as required by operating airline.



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Any damage to the structure or lining of containerized or bulk holds may lead to specific loading limitation. Therefore, any damage must be reported. The Load Controller shall be informed accordingly.

O. Compartment Separator Nets

Compartment separator nets must be secured prior to all flight departures. Between offload and on load, compartment nets must be secure inside aircraft compartments and not hanging outside, to avoid clips and attachment points from striking fuselage (especially during adverse weather conditions) or inadvertently hooked on GSE and pulled out of the aircraft.

P. Cargo Door barrier Nets

Between offload and on load, compartment nets must be secured inside aircraft compartments and not left hanging outside. Cargo door barrier nets must be installed prior to flight departure to prevent cargo from shifting in flight and damaging or blocking the compartment door.

Q. Container/ Pallet Restraint System

All container stops and pallet locks required to secure containers and pallets must be extended and locked prior to any flight. This includes cargo holds that are empty, unless specified by the operating airline. A tactile check must be performed by checking the security of each lock with your hands.

R. Advance Loading Preparation**ORM 2.4.1**

Before loading commences, the load shall be assembled and checked against loading instructions.

1. Ensure ULD placards are properly filled out with the correct information
2. Ensure ULDs and all cargo are inspected and are fit to be loaded on the aircraft.
3. If possible, arrange ULDS on the ramp in order of on load.

S. Aircraft Ground Stability**HDL 2.2.1**

Loading or offloading may cause the aircraft to become unstable or could cause the aircraft to tip. Respect aircraft ground stability requirements during loading and offloading In general:

1. Offload aft holds before forward holds.
2. When loading, load forward holds before aft holds.

NOTE: For cargo aircraft, a tail support stanchion or nose tether may be required to be fitted during loading and offloading.

T. Wet Cargo

The Loading Supervisor must check if:

1. The wet cargo is properly packed and free of leakage.
2. The aircraft floor is properly protected from risk of spillage.



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U. Loading Procedure**HDL 1.8.5, 1.8.6, 1.8.7, 2.1.3,
2.1.4, 2.1.6, 2.3.1**

Before loading commences, verify the aircraft registration with the registration on the loading instruction report.

1. Ensure on load has been checked against LIR Weights, commodity, number of pieces, and destination. ULD numbers must also be cross checked.
2. Ensure special equipment (tie down straps, etc.) is available, as required.
3. Ensure LIR is received and understood by loading crew.
4. Before loading commences, carry out inspection of cargo compartments and restraint system. Report any defects to supervisor, the flight crew, and/or a representative as required by operating airline.
5. Prior to release for loading ensure that all originating and transfer hold baggage has been subjected to appropriate security controls
6. For cargo shipments, ensure the nets or tie down straps are tight and the load is secure.
7. Inspect ULDs for serviceability. Do not load damaged ULDs.
8. Items with directional handling labels should be loaded so that the labels will be visible during offload.
9. When loading pallets or containers make sure that the edges are either guided by the side rails or fit under the stops/locks/guides and that the height of the pallet allows for sufficient clearance in the door opening.
10. Check that the passage of the ULDs into their position is not obstructed by stops/locks/guides.
11. Ensure separator nets, fire barriers, door nets, pallet locks and container stops are installed and locked as required as the hold is loaded.
12. Keep count of bulk loaded baggage by compartment and destination.
13. Documents all changes to the load, inform load control and sign the Load Instruction Report.
14. Retain Load Control documents as per customer airline requirements.
15. Carry out load verification prior to finalizing the weight and balance.
16. Ensure that the aircraft loading information is accurate and documented. Transfer the aircraft loading information to the individual responsible for the calculation of the final weight and balance.
17. Once the aircraft has been loaded, a Load Instruction Report is:
 - i. Completed and certified by the supervisor responsible for aircraft loading.
 - ii. Communicated to Load Control
 - iii. Retained as per customer airline requirements.



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Transportation must be in accordance with the IATA Live Animals Regulation (LAR). Also check airline manuals, corresponding sections, for aircraft specific regulations, which may apply.

When transporting and loading live animals we must:

- i. **ENSURE** that each animal is loaded and secured into a suitable aircraft compartments as directed by the Loading Instruction Report.
- ii. **ENSURE** that each animal is separated from foods, dangerous good or other AVI that are natural enemies.
- iii. **ENSURE** that each animal is handled in a manner to minimize the waiting period.
- iv. **ENSURE** that live animals are not exposed to adverse weather or environmental conditions during transportation, loading and unloading.

W. Load Spreading

When the weight of item(s) to be loaded exceeds the maximum floor load per square meter or the maximum floor load per running meter of a compartment, the weight has to spread to prevent damage to the compartment floor. This applies to HEAs, but may also apply to smaller items weighing less than 150 kg. The item must be fully restrained (see example below).



CAUTION: Overloading can cause damage to aircraft frames and ribs and consequently can have serious implications for the safety of the aircraft.

The weight can be spread by making use of spreading wood, in which case:

1. The surface to support the weight will be enlarged.
2. The length will be enlarged.

The Load Agent or cargo will advise the spreading requirements for each item. The information will be notified on the LIR.

X. Perishable Items**HDL 2.2.2**

When perishable or temperature sensitive healthcare items are to be loaded on the aircraft, it is important to ensure that they are brought to the aircraft as close to departure as possible to minimize wait times, while still ensuring that the load does not delay other items that need to be loaded. Perishable items should also be protected from adverse weather conditions during transportation, loading and offloading. Incompatible perishables must be separated from each other.

- Perishables must be moved into storage (i.e., cooler, freezer) appropriate to the type, in accordance with the PCR.
- Pharmaceuticals must be moved into storage (e.g., cooler, freezer) appropriate to the type, in accordance with the TCR.



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Cabin Supplies are normally provided by the client airline, and can be found on board of the aircraft having been screened at another station or they arrive along with the catering supplies and are subject to the same screening process as the catering service. Alternatively, any inflight supplies intended for transport on a passenger flight must be subjected to appropriate security controls, in accordance with applicable local regulations, and then protected from unauthorized interference until loaded onto the

Z. Securing the Load

When securing a load in an aircraft, it must be secured such that:

- i) It shall not move during the flight, which could dangerously affect the weight distribution and balance of the aircraft
- ii) It shall not cause damage to the aircraft structure or other important parts of the aircraft.
- iii) It shall not cause damage to another load or become damaged itself.
- iv) In case of an emergency landing, neither passengers nor crew are injured by the load.

J.3.5 AIRCRAFT OFF-LOADING**A. Scaling Process**

If the Flight Crew experiences a handling irregularity on take-off, the Flight Crew may request aircraft scaling (weighing of all baggage and cargo on board) at the arrival station. Aircraft must not be offloaded when a scaling has been requested until the process has been initiated.

B. Identifying Shipments Requiring Specific Handling

Comply with any special handling requirements. Take care with fragile Items. All shipments requiring specific handling will be identified on LDM/CPM. Make sure that packages with directional handling labels are kept in the correct orientation (this way up, etc.)

C. Safety Precautions for Offload

1. Take care when handling heavy item. Use proper lifting techniques and ask for assistance if required.
2. Take care when placing items on belt loaders. Make sure they are stable and will not fall off.
3. Check ULDS during offload for damage, leakage and load stability.
4. Check for incorrectly loaded ULDs (locks not raised, locks or side rails overridden, etc.). Report any discrepancies found.
5. Take care of load has shifted during flight.
6. Containers can tip during movement because the base is smaller than the top, causing a high center of gravity.



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J.3.6 IN-PLANE LOADING SYSTEMS**AGM 2.6.1, 2.6.2, 2.6.3**

In-plane loading systems must be operated under the client airline's procedures. When operating an in-plane loading system for loading and offloading ULDs use the following guidelines:

- During Loading, **ENSURE** that ULDs are guided into position by the use of side rails and/or stops, locks or guides.
- **ENSURE** that ULDs have an unobstructed path into the desired position.
- **USE LOCKS OR STOPS** to prevent any high-speed impacts.
- **ENSURE** that the ULDs are of a type approved for use for the specific aircraft.
- **ENSURE** there are no protrusion or overhangs that can damage the aircraft cargo door or the interior of the aircraft.
- Once in place, **SECURE ULDs** by using the aircraft floor locks.
- If the in-plane loading system is unserviceable, **IMMEDIATELY NOTIFY** the client airlines prior to loading or unloading of the aircraft. **ENSURE** that load control is informed and that adjustments to the weight & balance have been made if ULDs are to be loaded with any missing locks or latches.

J.3.7 CARGO TRANSPORT (DRAYAGE)**HDL 1.8.9**

When cargo, mail and baggage has to move between the cargo facility and the aircraft apply the following procedures.

- **COMMUNICATE & COORDINATE TIME** of transfer, to minimize exposure of cargo to the elements.
- **RECEIVE DOCUMENTATION**
- **ENSURE** that ground service equipment is serviceable (tractor, dollies, carts)
- **INSPECT THE CARGO**
 - i) **ENSURE** that you are collecting the correct cargo
 - ii) **ENSURE** that the cargo is not damaged and has no signs of tampering.
 - iii) **ENSURE** all ULDs are serviceable.
 - iv) **ENSURE** there are no nets, ropes, straps or protective materials dragging on the ground.
 - v) **ENSURE** all built up cargo is safe to move and will not shift, roll or topple.
 - vi) **ENSURE** that all dollies are serviceable and ALL LATCHES, LOCKS/STOPS ARE ENGAGED to keep ULDs on the dollies.
 - vii) **ENSURE** that all loose cargo is securely stowed and all handling instructions are followed.
- Make every effort to **PROTECT CARGO** from adverse weather conditions.
- **DO NOT PULL MORE THAN 4 DOLLIES at a time.**
- **DRIVE SLOWLY**
- **MONITOR CARGO DURING TRANSPORT**
- **RE-INSPECT CARGO UPON ARRIVAL**
- **HAND OVER DOCUMENTATION**



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Customer airlines may have one the following type of Dangerous Goods Programs:

A. WILL NOT CARRY

This program states that the carrier will not carry dangerous goods, therefore the only requirement will be to identify any hazardous items that may be packed in passenger baggage. Any hazardous items

The TSA will screen, identify and notify the carrier if any dangerous goods are identified in checked baggage.

B. WILL CARRY (Passenger Aircraft)**HDL 2.1.2 2.3.1, 2.3.2, 2.3.4, 2.3.6**

This program states that the carrier will carry cargo and dangerous good on the same aircraft as passengers.

The TSA will screen, identify and notify the carrier if any dangerous goods are identified in checked baggage, this type of undeclared hazardous item are prohibited.

Dangerous Goods shall only be accepted at the cargo facility and MUST BE handled, secured or stowed in a manner that:

- **PREVENTS DAMAGE** to packages and containers during aircraft loading and unloading
- **PREVENTS INTERACTION** between packages in case of leakage.
- **PREVENTS MOVEMENT** that could change the orientation of the packages inside the aircraft.
- **IS LOADED** in accordance with the required Notice to the PIC (NOTOC)
- If shipment is labeled **CARGO ONLY - DO NOT LOAD** onto an aircraft carrying passengers.
- **ENSURE** that hazardous items are loaded in accordance with the limited restrictions specified by the customer airline in the IATA DGR manual.

See Dangerous Good Regulations (DGR) book for instructions based on the substance being transported.

If a Dangerous goods baggage or shipment appears to be leaking:

- i) **DO NOT LOAD** the shipment on to the aircraft. **NOTIFY SUPERVISOR & OPERATIONS**
- ii) **IMMEDIATELY NOTIFY** the relevant authorities and the customer airline for instructions.
- iii) if already loaded, **REMOVE FROM THE AIRCRAFT**
- iv) Conduct an evaluation to identify and prevent any other cargo or baggage that may have been contaminated.

The flight coordinator or lead agent responsible for loading the aircraft will sign a Notice to Captain(NOTOC) confirming or otherwise that:

1. There was no evidence of leakage from the package(s) or any leakage from the ULDs loaded on the aircraft.
2. The UN number on the NOTOC matches the shipment label.
3. The package or ULD is loaded in the designated position and secured.

NOTE: The NOTOC must be retained and the information distributed as per the customer airline's requirements.



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C. WILL CARRY (Cargo Only Aircraft)**HDL 2.1.4**

This program applies to carriers who operate cargo only aircraft and do not transport passengers. Some Hazardous Items are not allowed to be transported on passenger aircraft. CARGO ONLY shipments are identified by a black and orange label. These shipments must be accompanied by a NOTOC.

The flight coordinator or lead agent responsible for loading the aircraft will sign a Notice to Captain confirming or otherwise that:

1. There was no evidence of leakage from the package(s) or any leakage from the ULDs loaded on the aircraft.
2. The UN number on the NOTOC matches the shipment label.
3. The package or ULD is loaded in the designated position and secured.

NOTE: The NOTOC must be retained and the information distributed as per the customer airline's requirements.



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Date: 10/01/2021**J.3.9 UNIT LOAD DEVICES (ULD)****HDL 1.6.1, ORM 2.4.1**

ULD's can be divided into two groups:

1. Containers and
2. Pallets with or without nets

Each ULD must meet minimum technical specifications to ensure safe restraint of the load. These specifications are published in the IATA ULD Technical Manual.

NOTE: Only ULDs that comply with the requirements of the customer airline's W&B Manual shall be loaded onto the aircraft.

A. Identification/ Labelling of ULDs**ORM 2.4.2**

Identification: Each ULD has an IATA identification code allowing proper ULD control. The first three identify the type of ULD. The next four or five identify the inventory number and the last two identify the airline or pool that owns the container. For example:

Type of Container	Inventory Number	Airline/Pool
AKE	12345	1A

All ULDs must be identified with container/pallet tags when load. The pre-printed letters (in boxes) indicate the specific application of the tag.

- Each tag must be fully completed.
- One tag must be placed in the tag holder of a container.
- A cross- check must be performed during the loading of the ULDs. The following identification numbers must always be checked to ensure they correspond with each other.
 1. ULD number shown on the LIR
 2. ULD number shown on the ULD identification tag
 3. ULD identification number printed or stamped on the ULD

B. Unit Load Device (ULD) Inspections**ORM 2.3.1, 2.5.2**

Unserviceable ULDs could:

1. Cause injury to employees.
2. Damage the aircraft structure.
3. Impact On Time Performance.
4. Damage ULD contents (Baggage and Cargo)
5. Affect Weight and Balance load requirements.

ULDs should be inspected for visual defects and damage when received or accepted and prior to being released for loading into an aircraft. In the event that a ULD has been identified as being damaged or not airworthy, the ULD must be tagged as unserviceable and stored in an area designated by the customer airline until it has been repaired, inspected and determined to be airworthy and placed back in service. A ULD Operational Damage Limits Notice will be attached.

C. ULD Handling and Storage**ORM 2.5.1**

Client Airline ULDs are stored in an overflow area designated by the airport or staged on dollies in the baggage make up area in preparation for the next flight operation in a manner that minimizes or eliminates the possibility of damage or loss.



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When the catering truck arrives the ground crew is responsible for the following.

1. **CHECK DRIVER ID AND SEAL RECEIPT.** If security is present this agent will perform this task.
2. Double check seal number against receipt if it is a match allow truck to service aircraft.
3. If seal on truck does not match the number written on receipt, open truck and visually search interior of truck and trolleys. Advise Ground Security Coordinator (GSC) and TSA.
4. **CHECK THAT THE LOAD IS PROPERLY SECURED.** (Carts tied down, all doors and shutters are closed.)
5. **ENSURE** that the catering vehicle does not approach the aircraft until the anti-collision lights have been turned off and there is not visual signs of damage on the vehicle.
ENSURE that the vehicle platform is perpendicular to the aircraft door sill.
5. Always **USE A GUIDE PERSON**. Make sure that trucks does not approaches nor reverses away aircraft without the use of a guide person. Always use company trained personnel as a guide.
6. **DO NOT** allow the catering truck to approach the aircraft in a raised position.
7. Leave a two inches gap between the rubber bumper of the stair and the aircraft fuselage.
DO NOT COME IN CONTACT WITH AIRCRAFT FUSELAGE.
8. Raise/ Lower vehicle body to the correct height.
9. **ENSURE** the vehicle body entry door is closed and latched.
10. **SHUT OFF ENGINE**
11. **ENSURE** Lower stabilizers (if necessary)
12. **ENSURE CHOCKS SET** with at least one chock at the front and one chock at the rear of the same wheel.
13. **ENSURE** that catering carts are pushed, not pulled on and off of the aircraft.
14. **DO NOT** stage catering carts on the vehicle platform.
15. **CONTINUALLY MONITOR** the clearance between the aircraft door and the vehicle platform.
Once Catering Service is complete, **PREPARE FOR BACK OFF.**
 - Secure Vehicle Loads
 - Close front vehicle body door
 - Check for FOD
 - Retract Guard Rails
 - Exit vehicle body, close and latch door
 - Raise Stabilizers
 - Lower Vehicle Body
 - Perform walk around
16. **ENSURE** that **BACK OFF ROUTE IS CLEAR.**
REMOVE CHOCKS
17. If necessary, **USE GUIDE PERSON** on the wing side. Stop if visual communication is broken.
18. Once the back of is completed, retrieve and stow the front chock.

**CAUTION:** In high wind conditions, **ENSURE** the following.:

- Increase distance between aircraft and catering vehicle.
- **DO NOT** Stow or Transport loose items on top of catering carts.
- **ONLY** push one catering cart at a time.
- **If WINDS ARE GREATER THAN 40kts, stop operations.**



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J.3.11 AIRCRAFT FUELING**HDL 1.2.14, 1.5.1**

Ultra does not perform aircraft fueling into plane. The ground crew is however responsible for the

1. Received fuel load from operations department and transmit to into plane fueling agent (when maintenance s not present
2. Obtain the final fuel load from Captain (if MTX is not present)
3. Check fueling agent ID for security purposes.
4. **ENSURE** fueling agent follows all the safety procedures, when approaching, hooking up to and departing from aircraft.
5. **ENSURE** that no GSE is blocking the exit route of the fueling vehicle.
6. Once the fueling is completed, collect the fuel ticket; if MTX is not present, and archive in flight file at the operations department.
7. **DO NOT SMOKE!**
8. In case of a spill, notify operations department and follow spill containment procedures.
9. When passengers are on board, embarking or disembarking **DURING FUELING** the area beneath such **EXITS MUST BE CLEAR** of GSE and/or other obstructions.
10. Refer to the operator airline's procedures for Notifications of persons on board the aircraft during fueling and aircraft evacuation procedures.

J.3.12 AIRCRAFT TOWING & PUSHBACK PROCEDURES**AGM 1.3.6, 1.3.12, 1.5.1**

- Prior to conducting a Towing or Pushback ensure that the equipment used for aircraft ground movement is suitable for the operation to be conducted. Check the chart on each pushback to verify that that the pushback is suitable to perform the intended operation.

Chart Example:

		AIRCRAFT TRACTOR	
		[MODEL]	
		RAMP CONDITIONS	
AIRCRAFT		DRY	WET
SMALLER			
ERJ 135/145			
MD80 / DC9			
B737 / B727			
A319 / A320			
A321			
B757 / B767			
B787 / B777			
A330/340/350			
B747			
COLOR KEY:			
RECOMMENDED			
USE CAUTION			
NOT ACCEPTABLE			
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• PUSHBACK / TOW TRACTOR COMPATIBILITY CHART

Use the chart below to determine tractor suitability for the intended operation.

PUSHBACK TRACTOR COMPATIBILITY CHART																																												
PUSHBACK TRACTOR	AIRBUS												BOEING					MCDONNELL DOUGLAS					OTHER / SMALLER																					
	A380	A350	A340-500/600	A340-200/300	A330	A300-600	A321	A320	A319	A318	A310	A330MRT	ATR42	ATR72	B777	B787	B757	B747	B737	B727	MD11	DC10/MD11	DC10 30/40	DC10 10	MD80	MD 82/88	MD90	DC9	R770	R85	R100	CRJ700	CRJ800	CRJ1000	DASH8 200	DASH 8 400	IL 76	IL96	ERJ135	ERJ140	ERJ195			
Hobart TA35A																																												
TD 300																																												
TPX 200 Towbarless																																												
JBT B250																																												
TLD TMX150																																												
GT-90																																												
MB-2																																												
S&S GT1628																																												
TMX 450-50																																												
TMX150-12																																												

RECOMMENDED

USE WITH CAUTION

NOT ACCEPTABLE / DO NOT USE

This chart is suitable for wet or dry ramp conditions. In wet conditions, perform movements at slower speed

- Before beginning a Tow or pushback, perform pre-departure procedures.

Perform Pre-Departure Walk-Around

- | | |
|-------------------------------------|---|
| a. FOD Clear | g. All Aircraft services panels closed. |
| b. Ramp Surface Conditions Suitable | h. ERA CLEAR of non-essential personnel |
| c. GSE Detached & Outside ERA | i. Cabin and Cargo Doors are flush with fuselage. |
| d. Power cables Clear | j. All Service Panels are closed and latched. |
| e. Passenger Loading Bridge Clear | k. No Visible Damage on aircraft. |
| f. Aircraft Path Clear | l. Report any abnormalities (damage, leaks, etc.) |

PREPARE FOR TOW OR PUSHBACK

- CONDUCT A FOD CHECK** on entire area removing all debris just prior to aircraft arrival.
- ENSURE** the stand surface condition is sufficiently free of ice, snow, etc.
- Make sure the aircraft path and ramp area is free of objects and obstacles which the aircraft may strike or endanger others due to jet blast effects.
- Make sure that marshalling staff, wing walkers and a jetway operator are present (if required).
- Disengage Jetway Auto-Level and Retract the Loading Bridge or Passenger Stairs from the aircraft.
- Establish Communication with brake operator / flight deck via headset or hand signals.



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Use the following procedures, when performing Aircraft Towing and Pushbacks.

A. CONNECTING THE PUSHBACK VEHICLE

AGM 1.3.7, 1.3.8, 1.4.1, 1.4.2, 1.4.3

The pushback vehicle is connected as follows:

1. Aircraft main gear chocks installed, nose gear chocks removed – if applicable;
2. Approach nose gear to centerline of fuselage;
3. When connecting a tow bar to the tractor, personnel face the tractor and, if possible, have both legs on the same side of the tow bar. (i.e., not straddling the tow bar).
4. Use a spotter/ guide person to assist in the final approach to nose gear;
5. **INSTALL NOSE GEAR STEERING BYPASS PIN**



DANGER: The bypass pin must be Labeled with the specific aircraft type(s) for which it can be used; Identified with a “Remove Before Flight” streamer; Checked regularly for proper technical condition, or as per manufacturer instructions.

NOTE: Each aircraft type has specific requirements for the bypass of the nose gear steering mechanism. Refer to the operating airline’s GOM for nose gear steering bypass pin details.

6. Verify that the bypass pin is installed correctly.

Tractor & Towbar:

- i) Connect tow bar to nose gear first.
- ii) Raise tow bar so that its head is at same height as the tractor connection.
- iii) Approach slowly until connection aligns and secure connection to tractor.
- iv) Raise tow bar wheels.
- v) Select “Neutral” or “Park” and set parking brake of tractor.
- vi) Remove Chocks prior to movement.

Towbarless Tractor:

- i) On final approach to aircraft, the tractor must be properly aligned and correctly positioned.
- ii) Position tow bar less tractor to standby for lifting and wait for approval from flight deck to lift, if applicable.
- iii) Select “Neutral” or “Park” and set the parking brake.
- vi) Remove Chocks prior to movement.



CAUTION: DO NOT REMOVE THE MAIN LANDING GEAR CHOCKS UNTIL; All GSE- with the exception of the boarding passenger stairs(s), GPU, PCA, and ASU is removed from the aircraft, the pushback vehicle is connected to the aircraft and the parking brakes of both the pushback vehicle and the aircraft are set.



NOSE GEAR PROTECTION AND STEERING ANGLES

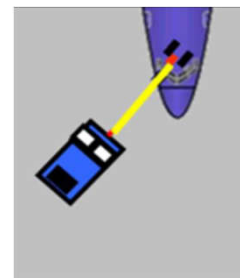
AGM 1.3.12

To protect the nose gear from damage, visual turning limit markings indicate the aircraft's maximum nose gear steering angles.

When transporting an aircraft with a towbar, it is recommended that the towing angle not exceed 45 degrees. Damage to the towbar or aircraft can result from towing or pushing an airplane while maintaining too sharp of an angle between the tractor and towbar. When making turns, a large sweeping radius should always be used. Turn limit will be reached when tow bar and markings in nose gear doors align, do not exceed these lines



CAUTION: DO NOT EXCEED 45 DEGREES TOWING ANGLE.



- There are markings on each aircraft indicating the maximum gear turn limit located on the (fuselage, nose gear, wheel well, etc). See examples below:



Example: Airbus



Example: Boeing



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B. AIRCRAFT PUSHBACK

AGM 1.3.5, 1.3.7, 1.3.8, 1.3.10, 1.3.11, 1.3.12, 1.3.14, 1.3.15, 1.3.16

- Pushback Requirements: All staff walking on the ramp **MUST REMAIN CLEAR OF HAZARD ZONES**, Aircraft Nose Gear throughout the pushback operation, the tractor's path; and Engine Danger Areas.
- If an aircraft is to be pulled forward after pushback and engines started, care and special precautions must be taken to reduce the risk of the aircraft's engine thrust causing damage to the nose gear and towbar when stopping the aircraft at completion of the maneuver. Special Precautions include gentle application of brake, engine at idle thrust, towing operation at lowest gear available.



CAUTION: When using towbarless tractor; **DO NOT** lift the aircraft when loading equipment and/or passenger boarding device is still connected to the aircraft.

- Ground Crew Responsibility: The ground crew for the departure will:
 - a) Ramp Lead Agent will be in charge of the entire pushback and assign responsibilities to ground crew, once clearance to begin pushback has been given by the flight crew. Only required personnel should be in the operating area during ground movement. Personnel involved in the operation will understand and agree with how communication will be performed and how the aircraft will be maneuvered.
 - b) **ENSURE PROPER TOWBAR & TRACTOR** is being used for the specific aircraft type;
NOTE: DO NOT leave paymover attached to an aircraft unattended while running.
 - c) **BE IN CONTINUOUS COMMUNICATION** with the flight crew
 - d) Inform the flight crew if ramp conditions are below the normal pushback conditions.
 - e) **ENSURE** that the nose gear **BYPASS STEERING PIN IS INSTALLED** prior to towbar connection to aircraft.
 - f) **CONNECT HEADSET**; Verify communication with the flight crew.
 - g) **KEEP FLIGHT CREW INFORMED** on progress of the ramp operation
 - h) Request permission & disconnect ground power unit (GPU) after verbal approval is received.
 - i) **CONDUCT PRE-DEPARTURE WALK AROUND.**
 - j) **SIGNAL "ALL CLEAR"** to pushback tractor driver and wing walkers that the aircraft brakes have been released and approval for pushback has been given by the flight crew.
 - k) **MONITOR HEADSET COMMUNICATION** and communicate with the flight crew as required.
 - l) Advise the flight crew if it is not safe to start an engine and stop the engine start.
 - m) Advise the flight crew to **SET AIRCRAFT BRAKES** at end of pushback.
 - n) **SIGNAL "BRAKES SET"** and give visual signal to **DISCONNECT TOWBAR** from the aircraft.
 - o) **DISCONNECT HEADSET and CLOSE ACCESS PANEL**
 - p) **REMOVE BYPASS PIN** from the nose gear and ensure the swing lever is returned to the proper position.
 - q) **CLOSE AND LATCH ALL ACCESS PANELS** and move to designated position to conduct final departure marshalling.
 - r) **ENSURE** verification of pin removal has been completed and **SIGNAL "ALL CLEAR"** to taxi.
 - s) Once flight crew has acknowledged **MOVE CLEAR** of the aircraft.



CAUTION: The flight crew (or brake operator) **MUST BE NOTIFIED IMMEDIATELY** in the event any connection between the paymover and the aircraft is lost during aircraft movement.



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- Paymover / Tractor Driver will: **AGM 3.1.11, 3.1.12, 3.1.13, 3.1.16, 3.2.2, 3.2.3, 3.2.4**
 - a) Align the tractor or tractor and towbar combination with the center line of the aircraft gear before the aircraft movement.
 - b) Completely raise the towbar wheels before the start of the aircraft movement (if used)
 - c) **STANDBY FOR CLEARANCE** to push communication from flight crew or responsible ground crew.
 - d) Select appropriate gear on tractor and **BEGIN MOVEMENT SLOWLY.**
 - e) Prior to aircraft movement, make sure that the parking brakes are released and the anti-collision lights are switched on (depending on the local airport regulations).
 - f) **START THE PUSHBACK** operation on a straight line.
 - g) Keep the maneuvering speed to a minimum, and apply the vehicle brakes gently.
 - h) Scan the apron during pushback, monitor clearances and wing walkers (if applicable) to ensure that the aircraft is moving clear of all obstructions. Be Prepared to Stop.
 - i) ENSURE during pushback that the steering turn limits are not exceeded. Notify flight crew if any limits are exceeded. Check with Client Airline's manuals for specific limits.
 - j) **MAINTAIN VISUAL CONTACT** and **ENSURE A SAFE DISTANCE** is maintained by any ramp crew from the nose wheel during the entire pushback.
 - k) If any ramp crew is seen not to maintain a safe distance from the nose wheel then Stop Pushback.
 - l) **SET BRAKES** once the pushback is complete.
 - m) MAINTAIN BRAKES until the release signal is received from the crew chief.
 - n) Wait for the flight crew to give the "AIRCRAFT BRAKES SET" signal.
 - o) **RELEASE PAYMOVER BRAKES** and put the gear selector in NEUTRAL after the aircraft brakes have been set to release pressure on the towbar.
 - p) **DISCONNECT THE TRACTOR** from the towbar, then disconnect the towbar from the aircraft.
 - p) Place the Paymover / Tractor in the aircraft's path and be visible to the flight crew after the towbar has been disconnected from the tractor.
 - q)
Remain in position until the headset operator has disconnected and is in view of the flight crew.
 - r) Drive tractor back to the terminal or an appropriate parking position.
REMINDER: "NO SEAT, NO RIDE"
 - For Open Ramp Departure:
 - a) Complete all pre-departure checks
 - b) Refer to departure communication section and follow required phrases of dialogue.
 - c) ENSURE all staff and equipment is clear of the aircraft and behind the ERA.
 - d) Position for marshalling in an area behind the ERA while being in clear view of the flight crew.
- AGM 3.1.11, 3.5.4**
- Maneuvering During Adverse Weather Conditions: During adverse weather conditions (fog, rain, ice or snow. etc.) visibility and traction will be affected. REDUCE SPEED as required by the present conditions.



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- Wing walker or other assist personnel must:
 - Be under the direction of the responsible ground crew at all times.
 - Use two (2) visible marshalling wands
 - Be positioned before and during movement of aircraft as follows:
 - Approximately 1 meter outboard of the wingtip.
 - In line with the rearmost main gear wheel
 - **MUST MAINTAIN VISUAL CONTACT** with the person responsible for the pushback/towing.
 - ENSURE** the aircraft movement path is clear of any obstructions.

At Arrival

- a) Two wing walkers will assume their position at the ramp boundary. Fuchsia line, to stop traffic as the aircraft approaches. As the aircraft is coming down the lead inline wing walkers will walk in to ensure that the marshaller can see then over the low visibility of the wings.
- b) Once the aircraft has come to a complete stop and beacon has been "turn off"
Wing walker will position the chocks in place (Inner most of the two main gear tires 2 and 4).
- c) Please insure to leave one inch separation between the chock and the tire for the chocks position in the front MLG tire.
- d) Connect GPU. Marshaler to give PIC proper signal for chocks inserted and GPU connected.
- e) Wing walkers will position safety cones. One at each wing tip, one at the nose and one at the tail end of the aircraft.

At Departure

- a) Two wing walkers will assume their position at the wingtips of the aircraft.
- b) Show "ALL CLEAR" signal to the Marshaller
- c) Once the aircraft begins to move maintain position relative to the aircraft while maintaining visual contact with the paymover driver and walk out with the aircraft.
- d) If any obstructions or hazards are seen, signal the tractor driver to STOP.
- e) Once the aircraft is in the departure position, maintain position relative to the aircraft until the towbar is removed, the headsets have been disconnected and the crew chief signals to return to the terminal.



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Date: 02/02/2022**C. AIRCRAFT TOWING PROCEDURE****AGM 1.5.1, 1.5.2, 1.5.3, 1.5.4**

The following requirements must be met to perform an aircraft tow:

- a) **ENSURE** hydraulic system pressure for aircraft braking and/or the brake accumulator is within required pressure range. Refer to the operating airline's GOM for each aircraft type for more detail.
- b) Ensure that any required electrical systems for towing are energized.
- c) Ensure all gear safety pins/sleeves are installed, and after tow, ensure all pins are removed and stowed.
- d) Make sure a qualified brake operator is in the cockpit.
- e) Establish communication with the brake operator by use of a headset.
- f) Make sure wheel chocks are positioned at the end of the maneuver, prior to disconnecting the towbarless tow tractor or towbar.



CAUTION: Inform the brake operator/flight crew and/or contact the maintenance department for technical inspection if you:

- observe any type of excessive fluid leakage;
- notice any signs of unmarked aircraft damage;
- observe any fault, failure, malfunction or defect which you believe may affect the safe operation of the aircraft for the intended flight.

TOWING MANEUVERING

The towing maneuvering procedure is similar for all aircraft types. The following minimum safety precautions and procedures must be followed prior to and during aircraft towing operations.

- a) Align the tractor and towbar combination with the center line of the aircraft before the aircraft movement.
- b) Completely raise the towbar wheels before the start of the aircraft movement.
- c) Prior to the aircraft movement, make sure that the parking brakes are released and the anti-collision lights are switched on.
- d) Wait for the authorization of the flight crew or brake operator before moving the aircraft.
- e) Start the pushback operation on a straight line
- f) Keep the maneuvering speed to a minimum, and apply the vehicle brakes gently.
- g) **DO NOT** exceed the towing speed limit as regulated by the towing equipment, aircraft and/or airport.
- h) Use the relevant apron lines as guidance during maneuvering to ensure safe obstacle clearance.
- i) Keep a minimum safe distance between vehicles sufficient in which to stop.
- j) Stop 50 m/55yd before a taxiway intersection, if a stop is required.
- k) Avoid sharp turns, which results in excessive tire scrubbing.
- l) Make all stops smoothly.
- m) When arriving at the allocated position, move the aircraft in a straight line for a few meters to ensure that the nose wheels are in the straight ahead position.
- n) Apply the tractor parking brake after a complete stop.



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TOWING LIMITS

Fuel and other loads can affect an aircraft's balance. To avoid "tail tipping" during towing, ENSURE that the actual center of gravity of the aircraft is forward of the critical center of gravity.

If the Brake Operator is an qualified customer airline mechanic, then confirm with him that the aircraft is safe to tow.

TOWING PREPARATION CHECKLIST**AGM 3.1.14, 3.1.15, 3.5.1, 3.5.4**

The following checklist is to be used in preparation for an aircraft tow.

ACTION	BRAKE OPERATOR	TRACTOR DRIVER
Apply the cockpit checklist for towing. Refer to the operating airline's GOM for details.	X	
Connect and test the interphone link.	X	
Insert the bypass pin.	X	X
Give permission to connect the towbar and tractor or towbarless tractor after applying the aircraft parking brake.	X	
Connect the towbar; first to the aircraft, then to the tractor.		X
Before connecting the towbarless tractor, ensure the aircraft main landing gears are symmetrically chocked.		X
Connect the tractor or towbarless tractor and set the parking brake.		X
Once all GSE has been cleared away from the aircraft, remove or check removal of aircraft chocks.		X
Switch on the external and anti-collision lights of the aircraft.	X	
Contact the Control Tower for clearance to start moving the aircraft (depending on local regulations).	X	X
After receiving the clearance, release the aircraft parking brake.	X	
Give clearance to the Tractor Driver to start moving the aircraft.	X	
Request confirmation from the Brake Operator that the aircraft parking brake has been released.		X
Conduct tow.		X



CAUTION: When towing in ice and snow, **MAINTAIN REDUCED SPEED** particularly when entering a turn and **AVOID STOPPING MOVEMENT** in a turn, to the extent possible.



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TOWING COMPLETION

The following checklist is to be used at the end of an aircraft tow.

ACTION	BRAKE OPERATOR	TRACTOR DRIVER
Set tractor parking brake.		X
Request Brake Operator to set the aircraft parking brake.		X
Inform the Control Tower that towing is completed and the frequency will be left (depending on local regulations).	X	X
Set the aircraft parking brake and check the pressure. Inform the Tractor Driver: PARKING BRAKE SET, PRESSURE CHECKED.	X	
Chock the aircraft main landing gear.		X
Switch off the external and anti-collision lights of the aircraft.	X	
Inform Brake Operator: AIRCRAFT CHOCKED.		X
Request permission from Brake Operator to disconnect the towbar or towbarless tractor.		X
Give permission to disconnect the towbar or towbarless tractor.	X	
Disconnect the towbar or towbarless tractor and remove the bypass pin.		X
Chock the aircraft.		X
Inform: TOWBAR/TRACTOR DISCONNECTED.		X
Release the aircraft parking brake and inform: PARKING BRAKE OFF.	X	
Check and inform: AIRCRAFT STABILIZED.		X
After permission from the Brake Operator, shut down and disconnect the tractor GPU.		X
Install and connect a GPU.		X
Remove and stow gear safety pins in the dedicated location.	X	



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INCIDENTS DURING TOWING**AGM 3.1.1, 3.5.3**

If an incident occurs during an aircraft towing, follow the instructions in the table below.

BRAKE OPERATOR	TRACTOR DRIVER
<i>VHF Communication Failure</i>	
	a) Stop aircraft/tractor set immediately b) Apply tractor parking brake. c) Advise Towing Regulation and wait for assistance.
<i>Tractor/Paymover Failure</i>	
a) Inform ATC b) Apply Parking Brake c) Listen to VHF and wait for Assistance.	a) Stop aircraft/tractor set b) Inform ATC (Towbarless Operation) c) Apply tractor parking brake. d) Chock the aircraft e) Listen to VHF (Towbarless Operation)
<i>Coupling Break Off</i>	
a) Brake the assembly by stepping on both brake pedals progressively. b) As soon as the aircraft is at a standstill, apply the parking brake before releasing the pedal.	a) Notify Flight Deck to stop Aircraft Movement. b) DO NOT apply tractor brakes. c) Follow the aircraft path attentively and stop the tractor according to the aircraft position. d) Chock the aircraft
<i>Tractor Fire</i>	
a) Inform ATC b) Apply Parking Brake	a) Inform the Brake Operator b) Stop aircraft/tractor set immediately c) Move tractor away as rapidly as possible d) Fight the fire, using the fire extinguisher. e) Chock the aircraft
<i>Aircraft Fire</i>	
a) Inform ATC b) Apply Parking Brake c) Fight Fire with the on board fire extinguisher. d) Evacuate the aircraft using on-board means, if required.	a) Stop aircraft/tractor set immediately b) Move tractor away as rapidly as possible c) Chock the aircraft
<i>Accident with Other Aircraft or Vehicle</i>	
a) Contact the Control Tower stating position and nature of trouble. b) Listen to VHF and wait for assistance.	a) Stop aircraft/tractor set immediately b) Apply tractor parking brake. c) Advise Towing Regulation d) DO NOT unload or disconnect the aircraft e) Chock the main landing gear.



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Date: 02/02/2022**J.4 AIRCRAFT ARRIVAL & DEPARTURE**

Planning and Preparation for the handling of an aircraft is key to safely and successfully providing ground handling services within the prescribed schedule. This section outlines the steps necessary to completing an arrival and departure. See specific procedure for additional detail.

J.4.1 COMMUNICATION**AGM 3.1.4**

Open Communication and awareness during ground handling operations are the most important components of conducting safe and efficient services for our client airlines. If you observe any unsafe conditions, report it to your supervisor immediately.

In order to standardize communication between ground staff members and between ground staff members and flight crew the following hand signals have been defined:

- A. Guide Person Hand Signals - to be used by a specific guide person to safely guide a GSE operator who is approaching an aircraft or pulling away from an aircraft or operating in low visibility conditions.
- B. Marshalling Hand Signals - to be used by ground staff to assist the flight crew during maneuvering of the aircraft and engine starting.

J.4.2 AIRCRAFT / GSE MARSHALLING**AGM 1.2.1, 1.2.2, 2.1.1, 2.1.2, 3.1.4**

The person giving hand signals must:

- **WEAR VISIBLE SAFETY VEST**
- When the vision of the GSE operator is or might be restricted, the movement of GSE operated in close proximity to the aircraft, is directed by one or more guide persons.
- **STANDARD MARSHALLING SIGNALS** are utilized by the guide person(s). The guide person(s) is/are positioned so that clearance from the aircraft, other equipment, vehicles or facilities can be accurately judged, and signals can be visually communicated to the GSE operator.
- If visual contact with the guide person is lost, the GSE operator stops movement of the GSE immediately. Operation would resume once visual contact is re-established.
- Illustrations of standard hand signals are displayed in the next section (MARSHALLING SIGNALS).
- Additional Guidelines for Giving and Receiving Hand Signals include:
 - i) The marshaller should position his/her hands well over his head or to the side of the body so the vehicle driver can easily understand the signal.
 - ii) The marshaller should always face the vehicle when it is being driven forward and reposition himself if he loses sight of the vehicle driver.
 - iii) The vehicle driver must stop if he loses sight of the marshaller or believes the vehicle is too close to the aircraft.
 - iv) When guiding from behind, the marshaller shall be visible to the driver but not directly in the path of the vehicle.
 - v) The marshaller should check the ramp area through which he walks to ensure that there are no obstructions.
 - vi) The marshaller must monitor the vehicle's path.
 - vii) The marshaller must use bright or lighted wands when visibility is reduced.



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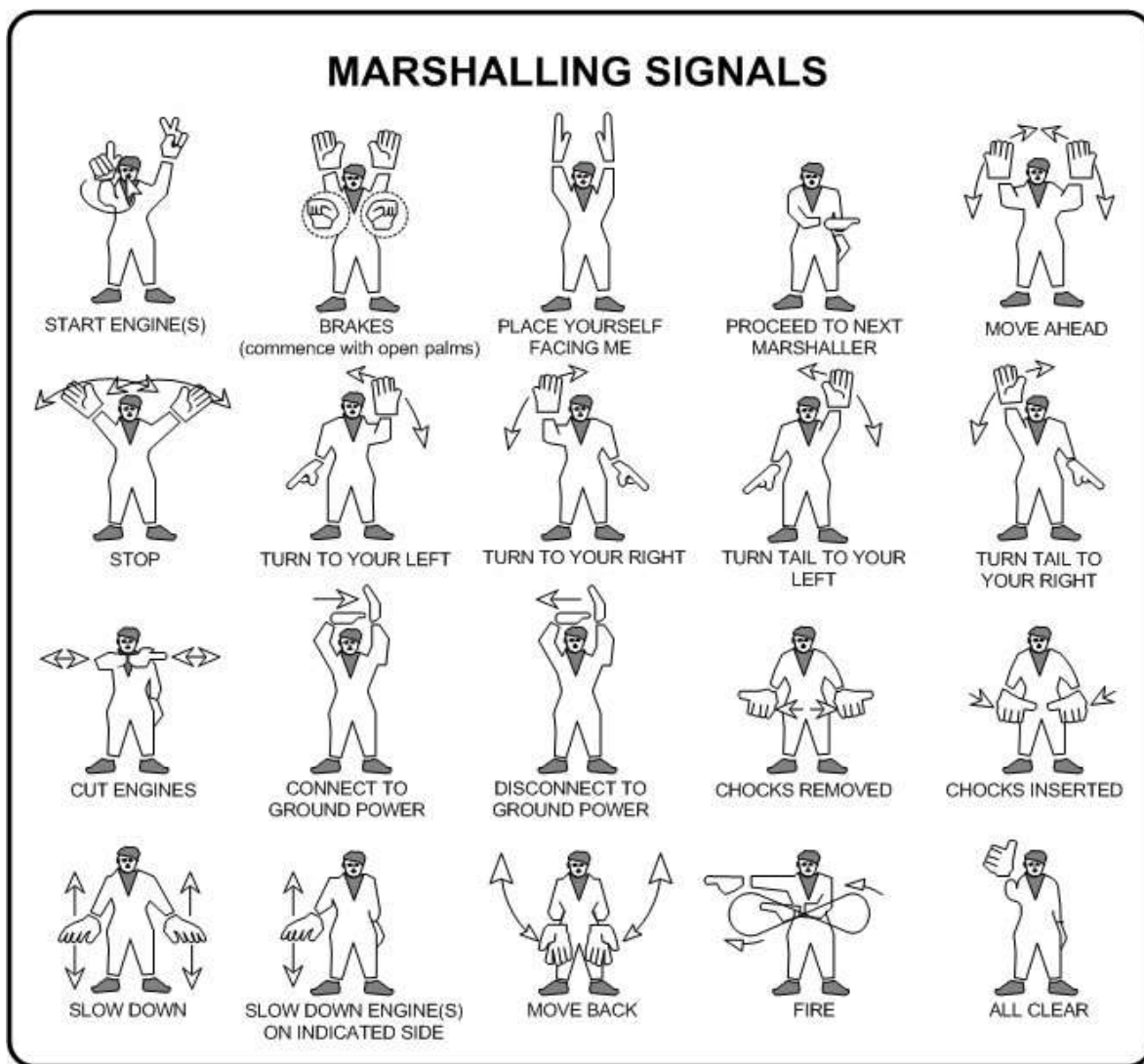
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J.4.3 MARSHALLING SIGNALS**AGM 1.2.1, 1.2.2**

The hand signals described are to be used for the marshaling and guidance of Company aircraft, and for communication between the flight crew and the ground crew when a service interphone is not available.

The aircraft handler, Company maintenance representative or qualified designee responsible for ramp operations must brief ramp personnel regarding the marshaling plan and any adjustments made to routine operations.

The pushback/ tow coordinator is responsible for the safe towing and pushback of the aircraft. The PIC has the ultimate responsibility for safe movement during taxi. The wing-walker uses hand signals to guide or assist the pushback/tow coordinator and/ or marshal in ensuring safe passage of the aircraft.





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J.4.3.1 GUIDE AGENT FOR GSE

- A.** The Guide Agent is an Employee who alerts a driver of obstacles when approaching/backing off from aircraft or equipment. A Guide Agent is required any time standard operating procedures cannot be safely followed while driving near aircraft and/or when the driver is uncertain of clearance.



WARNING: If the Guide Agent is not present or visible, **STOP** driving and inform the supervisor.

The following conditions must be adhered to when working with a Guide Agent:

1. Guide Agent must be a trained Ultra Employee
2. The Guide Agent must position themselves in a manner in which he/she can maintain visibility of the position of the GSE and its proximity to the aircraft in order to ensure that the GSE does not touch the aircraft.
3. The Guide Agent will, at all times, guide the GSE up to and away from the aircraft. The Guide Agent will also guide the GSE in congested areas.
4. Even though the driver and Guide Agent share the responsibility of ensuring safe ramp operations, the driver is ultimately in charge of the GSE's operation.
5. The Guide Agent should be careful not to give misleading or misdirected hand signals.
6. Should the Guide Agent give hand signals that do not follow standard operating procedures, the driver must stop immediately and inform the Guide Agent of the proper procedure.
7. Do not approach or back away from an aircraft with the Guide Agent in the truck.
8. The Guide Agent stands behind/in front of and off to the side of the GSE at a reasonably safe distance.
9. The Guide Agent must be in clear view of the driver and where the Guide Agent can view all obstacles.
10. Standing on the ladder or steps and giving signals is unacceptable.

CONDITIONS FOR USING HAND SIGNALS

The person giving the hand signals must:

- a. Use only approved hand signals.
- b. Wear a high-visibility vest.
- c. Maintain the same role throughout the procedure.
- d. Keep in constant visual contact with the other ground staff and flight crew throughout the procedure. If visual contact is lost, STOP and re-establish visual contact immediately.
- e. Remain clear of the intended pathway of the vehicle/aircraft where possible.



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Date: 02/02/2022**B. GSE GUIDE APPROACH PROCEDURE**

1. Get GSE Operator's Attention
2. Ensure that you are in a position in which you can Maintain constant visual contact with the GSE Operator; and Maintain a vantage point in which you can see the distance between the all parts of the GSE and the aircraft. Do not stand in a position where you can get caught between the GSE and the aircraft.
3. With visual contact established with the GSE operator, motion with hand signals to slowly begin movement using the approved IATA hand signals.
4. If necessary, indicate lift of GSE to appropriate height. (e.g. belt loader, high lift truck.
5. Use hand signals to motion for any necessary adjustments. (Left, Right, Straight)
6. Conduct three (3) **SAFETY STOPS** as required
6. When the GSE is almost in position, Motion to indicate the remaining distance.
7. Once the GSE is in position signal for the operator to STOP. Follow the company's **NO TOUCH POLICY.**
8. Insert chocks, Engage stabilizers.
9. Signal to Stop Engine.
10. Connect appropriate service apparatus.

C. GSE GUIDE REMOVAL PROCEDURE

1. Disconnect appropriate any and all service apparatus. (Hoses, Cables, etc.) Close Cabin Door(s).
2. Ensure that the exit path is clear.
3. Get GSE Operator's Attention
4. Ensure that you are in a position in which you can Maintain constant visual contact with the GSE Operator; and Maintain a vantage point in which you can see the distance between the all parts of the GSE and the aircraft.
5. Disengage stabilizers, Remove chocks,
6. Signal the ALL CLEAR, to Start Engine.
7. With visual contact established with the GSE operator, motion with hand signals to slowly begin movement.
8. If necessary, indicate lower of GSE to appropriate height. (e.g. belt loader, high lift truck.)
9. Use hand signals to motion for any necessary adjustments. (Left, Right, Straight)
10. When the GSE is clear of the aircraft , Motion to indicate ALL CLEAR to exit from the ERA.
11. Close and Secure any open Cargo Doors, access panels or hatches.



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Departure communication outlined in this section is a basic standard for both pushback and open ramp (taxi out) departures. Refer to the operating airline's GOM for specific requirements in their departure communication which may vary. This specific dialogue does not forbid the exchange of additional important information between flight crew and ground staff using non- standard phraseology.

In most cases, communication between the ground crew and the flight crew will be done via a headset plugged into the communication port on the aircraft. If a headset is not available or the communication system is inoperable, then use conventional hand signals for the departure communication.

Engine Fire

- The flight Crew normally detects an engine or APU fire and will take action using the engine fire extinguishing system. However, alert the flight crew immediately via the headset if flames are noticed from the engine or engine pylon.
- In the event that a headset is not available, the appropriate "Fire" hand signal must be used. (Refer to the Marshalling Hand Signals section in this chapter)

Tailpipe/Exhaust Fire

- If you notice flames from the engine tailpipe during engine starting, alert the flight crew immediately, as such a fire might not be detectable via temperature sensors and/or fire warning systems in the aircraft.



CAUTION: DO NOT FIGHT ENGINE FIRES WITH FIRE EXTINGUISHERS when the flight crew is in the flight deck. The flight crew will take all necessary action.

Loss of headset communication

- In the event that there is a loss of communication connection between the tractor and the aircraft during a tow or pushback, **IMMEDIATELY NOTIFY THE FLIGHT DECK.**

If headset communication is broken:

1. **IMMEDIATELY NOTIFY THE FLIGHT DECK.**
2. Attempt to re-establish headset communication by verifying the headset connection points and/or replacing the headset with another, if available.
3. If headset communication is still not possible, use hand signals to communicate with the flight deck throughout the pushback process.



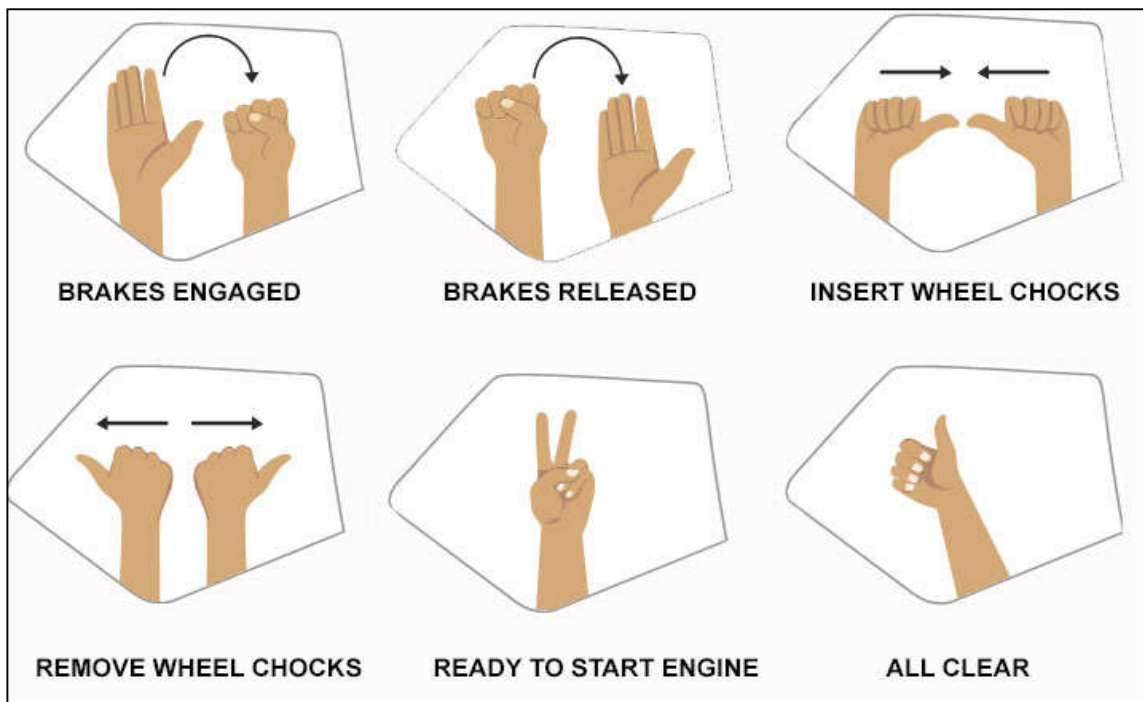
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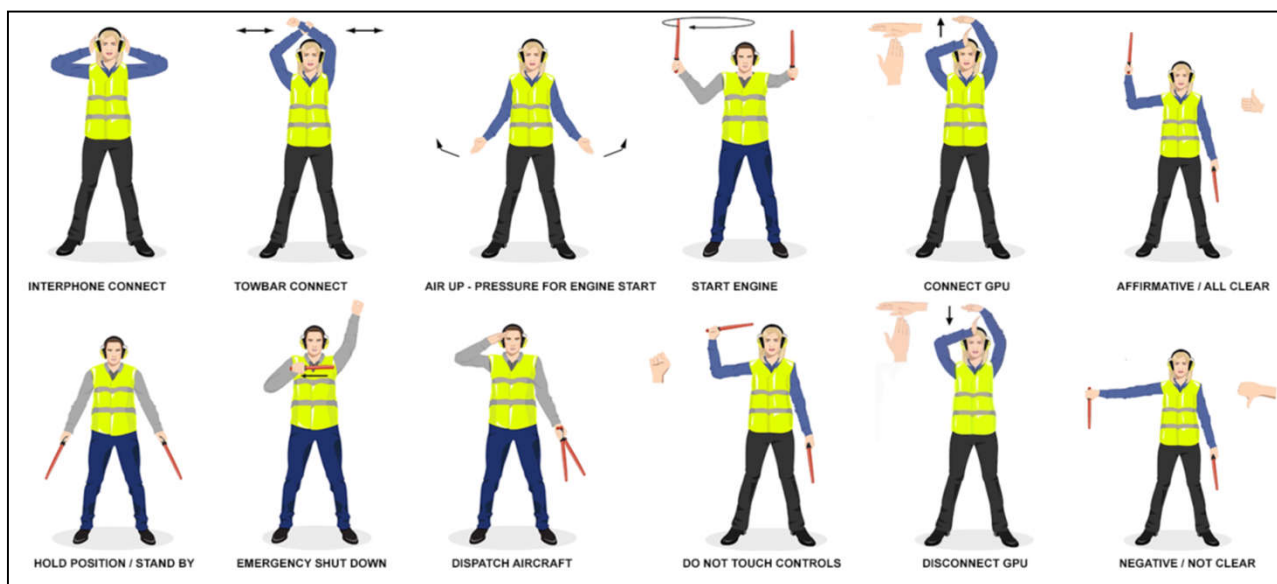
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a. Technical / Servicing Hand Signals (Flight Crew to Ground Crew)



b. Technical / Servicing Hand Signals (Ground Crew to Flight Crew)





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COMMUNICATION DURING PUSHBACK

The following items should be communicated between Ground Staff and the Flight Crew.

PHASE	TASK	GROUND STAFF ACTION
Departure Preparation	GPU Removal	When instructed by flight crew, remove GPU.
	Towbar / Tractor Connection	i) Get confirmation that the aircraft's parking brake is set. ii) Advise flight crew when the bypass pin is inserted. iii) Connect the Towbar. iv) Connect the Towbarless Tractor.
	Chock Removal	i) Get confirmation from flight crew that aircraft parking brakes are set ii) Remove Chocks
	Pre-Departure Check	Advise the flight crew that the pre-departure check has been completed or communicate any discrepancies.
Engine Start (Only)	Starting Engines	When requested by the flight crew, advise when the engines may be started and start sequence.
	Air Start Unit (ASU)	When requested by the flight crew, signal ASU operator to supply the required pressure.
Pushback & Engine Start	Brakes	Get confirmation that aircraft's parking brakes have been released.
	Movement of the aircraft (PUSHBACK)	Get permission from flight crew, to commence the pushback.
	Direction of push/nose	If applicable, ask in which direction the aircraft should the nose point after pushback.
	Engine Start	When requested by the flight crew, advise when the engines may be started and start sequence.
Pushback Completed & Engine Start Completed.	Towbar / Tractor Disconnect	i) Get confirmation that the aircraft's parking brake is set. ii) Disconnect. iii) Remove the steering bypass pin.
	Headset Removal	i) Get permission from flight crew to disconnect the headset. ii) Advise flight crew to hold position and wait for visual signal at left/right of the aircraft.
Departure	"ALL CLEAR" SIGNAL	i) Ensure verification of pin removal has been completed. ii) Give the "ALL CLEAR" signal when the path of the aircraft is clear of all obstacles. iii) Get acknowledgement of "ALL CLEAR" signal from flight crew.



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J.4.5 PASSENGER LOADING BRIDGE / JETWAY PROCEDURE (PLB)**HDL 1.4.2, 1.4.3, 1.4.4, 1.4.9**

Passenger Loading Bridges or Jetways are used at terminal gates to allow passengers to embark and disembark the aircraft to/from the terminal building.

ONLY AGENTS WITH CURRENT TRAINING FROM THE AIRPORT AUTHORITY MAY OPERATE JETWAYS.

1. Conduct a physical check of the bridge integrity and functionality.
 - Lights
 - Doors
 - Tires
 - Camera(s) (is operational)
 - Possible obstructions
 - Check that the PLB path is clear
 - Check for FOD
 - **IMPORTANT:** Prior to aircraft arrival and departure movement **ENSURE** that the Jetway is fully retracted and that the aircraft path is clear from obstructions.
 - **NOTE: ONLY THE JETWAY OPERATOR** should be in the bridgehead. If vision is restricted use a guide person.
 - **REPORT ANY MALFUNCTIONS TO THE AIRPORT AUTHORITY.**

To Attach the Jetway**HDL 1.4.5, 1.4.6, 1.4.7 , 1.4.8**

2. Turn Master Switch to "Operate"
3. Push power button
4. Using joystick check directional controls
5. Set elevation, by pressing Vertical directional control on the UP or DOWN direction.
6. Set cab inclination by pressing inclination control Left or Right.
7. Using directional JOYSTICK slowly approach designated aircraft door (L1 or L2)
NOTE: Do not move PLB towards aircraft until the ANTI-COLLISION LIGHT (beacon) has been turn off.
8. **MOVE JETWAY SLOWLY** until the bridge safety bar just touches the aircraft. **AVOID** any protruding aircraft components.
9. Position PLB close to and aligned with the aircraft door leaving a two (2) inches gap between the rubber bumper and the fuselage. PLB should be one(1) to two (2) inches below door sill, depending in aircraft door configuration.
10. Lower the Canopy
11. Once the Jetway is in position **ENGAGE AUTO LEVEL**. Turn master switch from OPERATE to AUTO LEVEL. **NOTE:** Not setting auto leveler could damage the aircraft since the PLB will not move up and or down as the aircraft settle when deplaning and or embarking.
12. **SECURE PLB CONTROLS**, if possible.

To Remove the Jetway

13. **PLACE PLB SAFETY BARRIERS**, if available.
14. Once aircraft door has been closed, Turn master switch to OPERATE
15. Raise the Canopy.
16. Open side door and visually check that there is not obstruction on the PLB intended path.
17. Using JOYSTICK move PLB away from aircraft.
18. Once aircraft has departed from the gate, position PLB back on storage "BOX"
19. Lower rolling door and turn all cabs light off
20. Turn master switch to OFF



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Date: 02/02/2022**J.4.6 PRE-ARRIVAL PROCEDURE****AGM 1.1.2**

1. Obtain Flight Information, Arrival Time and Gate/Position from Operations Department (Email, Phone Call, Supervisor, Online Portal)
2. Confirm personnel scheduled and available for aircraft arrival.
3. **At ETA - 20** have crew report to the Arrival Gate/Position.
4. **CONDUCT A FOD CHECK** on entire area removing all debris just prior to aircraft arrival.
5. **ENSURE** the stand surface condition is sufficiently free of ice, snow, etc.
6. Make sure all required Ground Service Equipment (GSE) is available and serviceable, and is positioned well clear of the aircraft path. Outside the Equipment Restraint Area (ERA).
 - 6 Marshalling Wands
 - 4 Cones
 - 4 Chocks
 - 2 Tugs/Tractors
 - 2 Belt Loaders
 - 1 Ground Power Unit
 - 1 Air Condition Unit
 - 2 ULD Loaders (Wide-Body)
 - Ramp Crew
 - Jetway Operator
 - Headset
7. Make sure the aircraft path and ramp area is free of objects and obstacles which the aircraft may strike or endanger others due to jet blast effects.
8. Make sure that marshalling staff, wing walkers and a jetway operator are present (if required).

J.4.7 GENERAL ARRIVAL PROCEDURE**AGM 1.1.3, 1.1.4, 1.1.5**

1. Marshall Aircraft into position. (See Marshalling Procedures)
As Engines are spooling down and the anti-collision light has been turned off:
2. Position Chocks and Safety Cones and Notify Flight Crew.
3. Arrival Walk-Around - Inspect for damage on the following parts of the aircraft: (Cargo Doors, Access Panels, Fuselage, Engine Cowlings, and Passenger Doors)
***Note:** If any damage is found, report it immediately to the supervisor and do not approach the aircraft with any GSE in the area where the damage has been found.*
4. Position Loading Bridge or Stairs. Once the Jetway is in position and the auto-level is on, knock on the passenger door to signal a trained crew member to open the aircraft door.
5. Disembark Passengers (Aisle Chairs, Wheelchairs, VIP Passengers)
6. Give "ALL CLEAR" for GSE to approach aircraft. Follow the appropriate procedure for each type of GSE.
7. Position and Attach any auxiliary ground service equipment, such as GPU or Air Conditioning.
8. Open Cargo Doors and Offload Baggage on to carts or dollies to be taken to baggage claim.
9. Clean Aircraft according to carrier procedures and standards. For International Arrival: Follow all "Regulated Garbage" procedures.
10. Communicate - Block Times & Fuel Remaining, and Service Requests to Operations Department.
11. Lavatory Service - See Lavatory Service Procedure.
11. Complete Arrival Procedure - Prepare for Departure, Prepare for Towing, of Secure Aircraft if arrival is a Terminating Flight.



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Date: 02/02/2022**J.4.8 PRE-DEPARTURE / AIRCRAFT SERVICE PROCEDURE****HDL 1.3.6**

1. Obtain/Confirm Flight Information, Departure Time and Gate/Position from Operations Department (Email, Phone Call, Supervisor, Online Portal)
2. If the aircraft is not parked at the appropriate Gate/Position, then check with operations and if necessary, tow the aircraft to the appropriate Gate/Position.
3. Make sure all required Ground Service Equipment (GSE) is available and serviceable for the weight and type of aircraft, weather and surface conditions, and is positioned well clear of the aircraft path. Outside the Equipment Restraint Area (ERA).
 - 6 Marshalling Wands
 - 1 Pushback Tractor
 - 1 Appropriate Towbar
 - 1 Gear Bypass Pin
 - 2 Tugs/Tractors
 - 2 Belt Loaders
 - 1 Ground Power Unit
 - 1 Air Condition Unit
 - 2 ULD Loaders (Wide-Body)
 - 1 Air Start Unit, if necessary
 - Ramp Crew
 - Jetway Operator
 - Security Personnel as required
 - Headset
4. **Lead Agent: ASSIGN RAMP CREW THEIR ROLES.** Only persons required to perform operations should be present.
5. **SECURE AIRCRAFT ACCESS** - If required by TSA for assigned operation. Control access to the aircraft by placing a guard at the cabin door during aircraft handling operations. On the ramp, Check ID and screen any ramp agent entering or exiting the cargo hold area.
6. Service the Aircraft - Fuel Uplift, Catering, Water Service, additional Lavatory Service, if necessary
7. Complete Load Plan and Retrieve baggage from baggage make up area and outbound cargo from cargo warehouse.
8. **LOAD THE AIRCRAFT - Ramp Crew:** Position Belt / ULD Loaders and load the aircraft according to the load plan and by following company and carrier aircraft loading procedures. **Security:** Conduct positive bag match (bingo sheet or scanner system).
9. Perform Aircraft Search according to TSA Regulations.
10. Position Pushback Tractor and connect appropriate towbar to both the tractor and the aircraft for departure.
11. Paperwork - Confirm Flight Folder with all required documents and information has been delivered to the Flight Crew. (GenDecs, Weather, Flight Plan, Load Plan, etc.)
12. Confirm Weight & Balance with the Pilot in Command (PIC) and complete baggage/cargo loading.
13. Confirm that all Aircraft Servicing has been completed. (Fuel, Water/Lav, Catering, Maintenance)
14. Confirm that Flight and Cabin Crew is ready for boarding and board passengers.
15. Once Boarding and Loading are complete, close cargo bins, close aircraft doors.
16. Carefully Remove all GSE from the aircraft and securely position outside of the ERA.
17. **PREPARE FOR DEPARTURE**
18. **CONDUCT A FOD CHECK** on entire area removing all debris just prior to aircraft arrival.
19. **ENSURE** the stand surface condition is sufficiently free of ice, snow, etc.
20. Make sure the aircraft path and ramp area is free of objects and obstacles which the aircraft may strike or endanger others due to jet blast effects.
21. Make sure that marshalling staff, wing walkers and a jetway operator are present (if required).
22. Disengage Jetway Auto-Level and Retract the Loading Bridge or Passenger Stairs from the aircraft.
23. **READY FOR DEPARTURE**



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J.4.9 DEPARTURE / PUSH BACK PROCEDURE

AGM 1.3.1, 1.3.2, 1.3.3, 1.3.4, 1.5.1

1. Perform Pre-Departure Walk-Around
 - a. FOD Clear
 - b. Ramp Surface Conditions Suitable
 - c. GSE Detached & Outside ERA
 - d. Power cables Clear
 - e. Passenger Loading Bridge Clear
 - f. Aircraft Path Clear
 - g. All Aircraft services panels closed.
 - h. ERA CLEAR of non-essential personnel
 - i. Cabin and Cargo Doors are flush with fuselage.
 - j. All Service Panels are closed and latched.
 - k. No Visible Damage on aircraft.
 - l. Report any abnormalities (damage, leaks, etc.)
 - m. Landing Gear Safety Pins Removed.
2. Connect Headset and establish communication with the cockpit crew.
3. **CHECK PUSHBACK VEHICLE CONNECTION**
4. Verify that **Wing Walkers** are in position. Wearing Safety Vests or Reflective Shirts.
5. **REMOVE CHOCKS, NOTIFY FLIGHT CREW, RECEIVE ACKNOWLEDGEMENT FROM FLIGHT CREW.**
6. **PUSHBACK** Aircraft from the gate to face airport taxi ways. See Pushback Procedure in this chapter.
7. SIGNAL START ENGINE #2
8. SIGNAL START ENGINE #1
9. REMOVE TOWBAR from the aircraft
10. CONFIRM "GO" from the crew.
11. DISCONNECT HEADSET - Disconnect headset and close communications access panel.
12. CLEAR PUSHBACK TRACTOR away from the aircraft.
13. **AIRCRAFT DEPARTURE**
14. Once push back has been terminated and the departure salute has been given to PIC all crew will return to gate.
15. Scan for FOD walk while walking back to the gate.
16. All GSE will be pick up and returned to its storage position.
17. Any Left Over Baggage (LOB) will be taken to Bagroom Makeup Area (BMU) to be logged and secured.
18. Notify Operations of Flight Departure.



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J.4.10 SECURE AIRCRAFT**HDL 3.2.2**

It is necessary to secure the aircraft according to customer airlines procedures. When and aircraft is to be closed and secured during a layover or overnight parking **ENSURE** the following:

Perform Pre-Departure Walk-Around

- | | |
|-------------------------------------|---|
| a. FOD Clear | g. All Aircraft services panels closed. |
| b. Ramp Surface Conditions Suitable | h. ERA CLEAR of non-essential personnel |
| c. GSE Detached & Outside ERA | i. Cabin and Cargo Doors are flush with fuselage. |
| d. Power cables Clear | j. All Service Panels are closed and latched. |
| e. Passenger Loading Bridge Clear | k. No Visible Damage on aircraft. |
| f. Aircraft Path Clear | l. Report any abnormalities (damage, leaks, etc.) |

- Place a tamper evident seal on Cabin Doors if necessary.
- **DO NOT LEAVE AIRCRAFT** if the above items have not been accomplished.

J.4.11 TAXI OUT DEPARTURE PROCEDURE.**AGM 2.2.1**1. Perform Pre-Departure Walk-Around

- | | |
|---|---|
| a. FOD Clear | h. ERA CLEAR of non-essential personnel |
| b. Ramp Surface Conditions Suitable | i. Cabin and Cargo Doors are flush with fuselage. |
| c. GSE Detached & Outside ERA | j. All Service Panels are closed and latched. |
| d. Power cables Clear | k. No Visible Damage on aircraft. |
| e. Passenger Loading Bridge Clear | l. Report any abnormalities (damage, leaks, etc.) |
| f. Aircraft Path Clear | m. Landing Gear Safety Pins Removed. |
| g. All Aircraft services panels closed. | |
2. Connect Headset and establish communication with the cockpit crew.
 3. **REMOVE CHOCKS, NOTIFY FLIGHT CREW, RECEIVE ACKNOWLEDGEMENT FROM FLIGHT CREW.**
 4. SIGNAL START ENGINE #2
 5. SIGNAL START ENGINE #1
 6. CONFIRM "GO" from the crew.
 7. DISCONNECT HEADSET - Disconnect headset and close communications access panel.
 8. **AIRCRAFT DEPARTURE**
 9. Once departure salute has been given to PIC all crew will return to gate.
 10. Scan for FOD walk while walking back to the gate.
 11. All GSE will be pick up and returned to its storage position.
 12. Any Left Over Baggage (LOB) will be taken to Bagroom Makeup Area (BMU) to be logged and secured.
 13. Notify Operations of Flight Departure.



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Revision: 7
Date: 02/02/2022**J.5 GROUND SERVICE EQUIPMENT (GSE) MANAGEMENT**

Safety is the top priority while conducting ground handling operations; therefore it is crucial that any ground service equipment utilized is maintained in good serviceable condition.

J.5.1 GROUND SERVICE EQUIPMENT (GSE) MAINTENANCE

Any ground service equipment utilized by the company to perform ground handling services shall be maintained under a maintenance program that ensures that GSE remains safe to operate and in good condition.

The company could outsource the maintenance of its ground service equipment. The company shall conduct periodic audits of any Ground Service Maintenance providers as prescribed in section C.4.5 of this manual in accordance with (ORM 3.6.2) to ensure that:

- Any GSE utilized in ground handling operations is maintained in accordance and/or guidance from the GSE manufacturer.
- Mechanics who are working on any GSE are properly training and have the appropriate knowledge to maintain, repair and return GSE to a safe and serviceable condition.
- Maintenance records are retained for a minimum period of one (1) year, unless otherwise required by applicable regulations or customer airlines.



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